

DSP Integrated Circuits

Chapter 4: Digital Filters

Prof. Jaeseok Kim

Communication SoC Design Lab.

Yonsei University

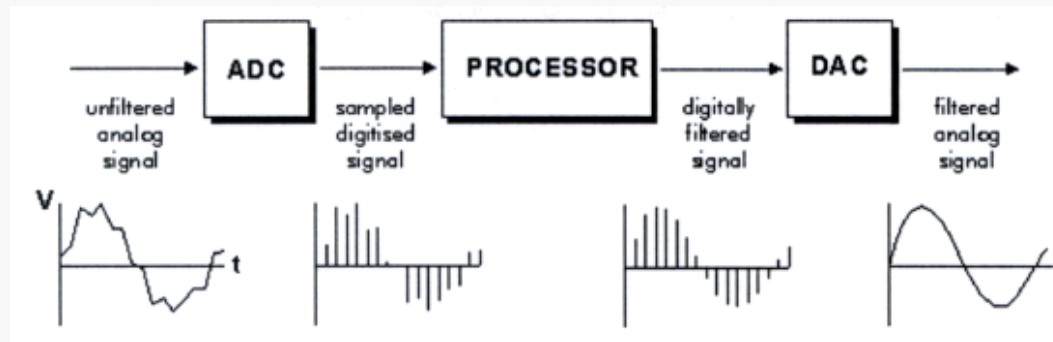
Contents

- ◇ Introduction to digital filters
- ◇ FIR filter characteristics and structures
- ◇ IIR filter characteristics and structures
- ◇ Multirate signal processing
 - Interpolator
 - Decimator
- * Design Example of Digital SAW filter



Introduction to digital filters

◇ Digital filtering of signals



◇ Applications of digital filters

- Digital Audio
- Telecommunications (noise cancellation or data compression)
- Frequency synthesizers
- Harmonic cancellers
- Correlators

Advantages of digital filters

1) stable & repeatable

- No drift with time or temperature
- R, L, C components are replaced by digital parameters

2) Adaptable

- Filter parameters are programmable & reconfigurable

3) High performance

- Meet demanding freq. response without phase errors

4) Predictable

- Design testable by simulation in S/W



Digital Filter Types

1) FIR (Finite-length Impulse Response) filters :

- Non-recursive (no feedback) type
- Guaranteed to be stable since there are no poles
- Have a linear-phase response
- Large amounts of memory and arithmetic processing needed

2) IIR (Infinite-length Impulse Response) filters :

- Recursive type
- Required filter order is much lower than FIR filter
- Analogy with analog filter design (Butterworth, Chebyshev, ...)
- Have stability problems

3) Lattice filter

- A type of digital IIR filter whose performance lies between FIR & IIR
- Provides greater stability than IIR



Use FIR or IIR ?

◇ Use FIR if

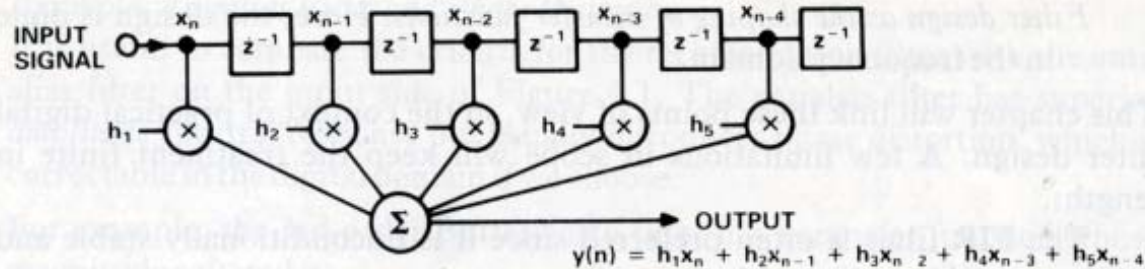
- $H(f)$ can be specified as a set of pass- and stop-bands
- Needs guaranteed stability and linear phase
- CAD techniques provide an interactive cookbook approach

◇ Use IIR if

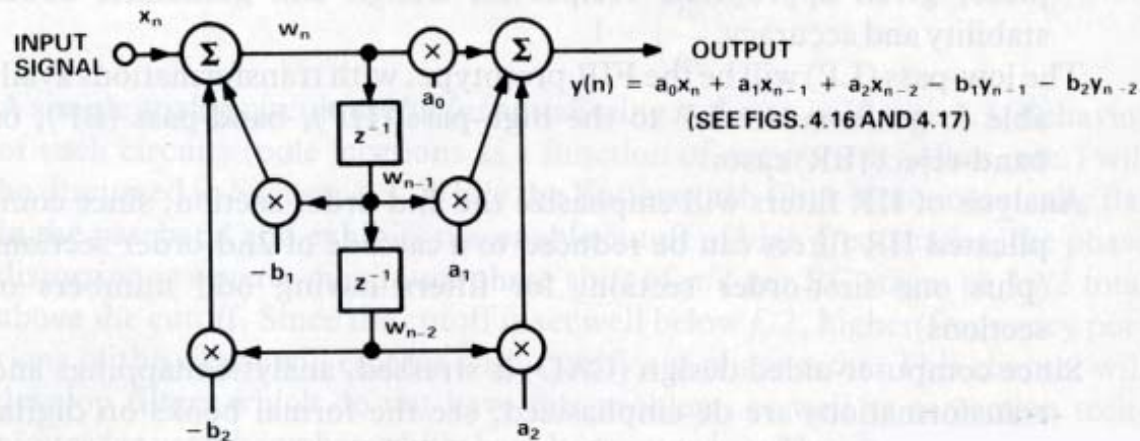
- The analytic form of $H(f)$ matters
(Ex. Formant filters for speech are modeled by poles & zeros)
- Easy conversion from analog counterpart filter



Block diagrams of FIR & IIR filters



a. FIR or finite impulse response.



b. Basic IIR or infinite impulse response.



FIR Filters

FIR filters

- ◇ FIR filter response of order N :

$$y(n) = \sum_{k=0}^{N-1} a(k) \cdot x(n-k)$$

- ◇ Filter transfer function :

$$H(z) = \sum_{n=0}^{N-1} h(n)z^{-n} = h(0)z^0 + h(1)z^{-1} + h(2)z^{-2} + \dots + h(N-1)z^{-N+1}$$

- ◇ Poles of transfer function are at the origin of Z-plane for FIR non-recursive algorithms
- ◇ Zeros can be placed anywhere in Z-plane, but most are located on the unit circle for better attenuation



Linear-Phase FIR filters (1)

- ◇ FIR filters, with coefficients that are either symmetric or antisymmetric, do not introduce waveform distortion because the phase shift is linear in frequency (constant group delay)

- $$H(f) = \sum_{n=0}^{N-1} h(n) \cdot e^{j2\pi nf}$$

- Center the sum at $n=0$ by making a change of variables, $n \rightarrow n - N/2$, which brings out a phase shift $e^{j\pi Nf}$, and express the exponential by its sine & cosine components

- $$H(f) = e^{j\pi Nf} \cdot \sum_{n=0}^{N/2-1} [h(n) \cdot \cos(2\pi nf) + jh(n) \cdot \sin(2\pi nf)]$$



Linear-Phase FIR filters (2)

- For the *symmetric* case,

$$h(n) = h(N-n), \quad n = 0, 1, \dots, N$$

$$\phi(H(f)) = -\pi Nf \rightarrow \text{Linear phase shift}$$

- For the *antisymmetric* case,

$$h(n) = -h(N-n), \quad n = 0, 1, \dots, N$$

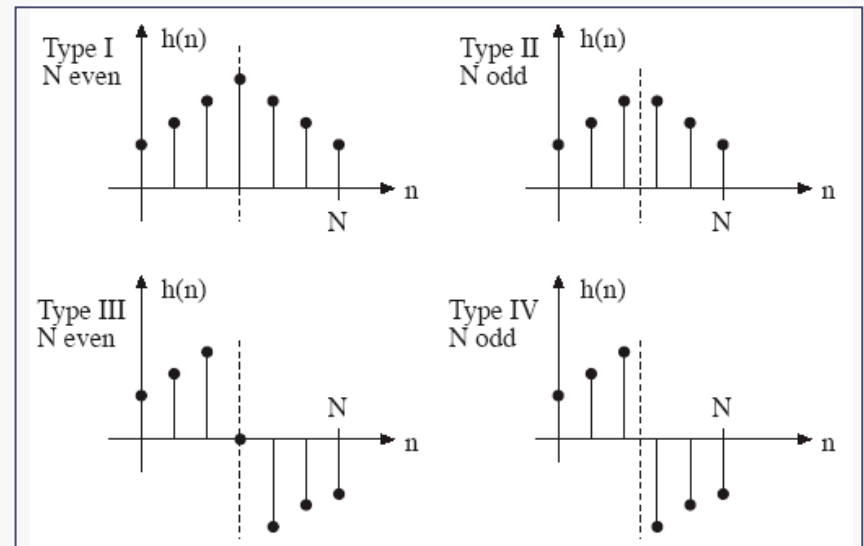
Phase shift of $H(f)$ remains linear

Type I: $h(n) = h(N-n)$, N even

Type II: $h(n) = h(N-n)$, N odd

Type III: $h(n) = -h(N-n)$, N even

Type IV: $h(n) = -h(N-n)$, N odd



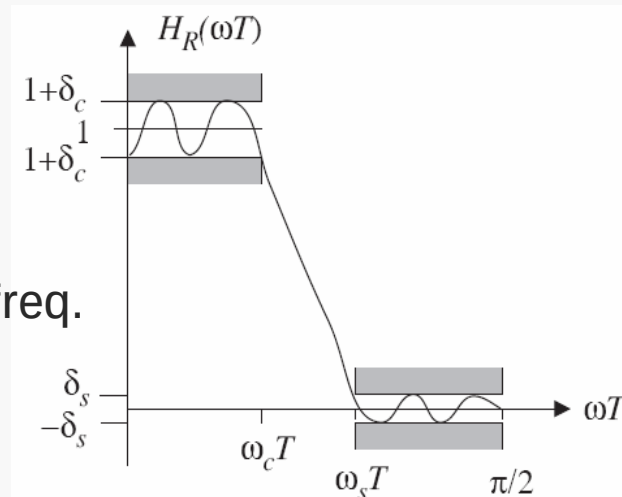
Design of FIR filters (1)

- ◇ FIR Spec. is expressed in terms of filter transfer function $H_R(\omega T)$ with $\omega_c, \omega_s, \delta_c$ & δ_s

* Where ω_c : passband freq., ω_s : stopband freq.

- ◇ The acceptable deviations are $\pm\delta_c$ & $\pm\delta_s$ in passband and stopband, respectively

- ◇ Numeric optimization procedures such as McClellan Parks-Rubiners program using Remez exchange algorithm are used to determine the coefficients in $H(z)$



Design of FIR filters (2)

- ◆ The length of impulse response for linear-phase filter that meets the spec. shown in Figure can be estimated by

$$N \approx -\frac{2}{3} \log_{10}(10\delta_c\delta_s) \frac{\pi}{(\omega_s - \omega_c)T} + 1$$

- Filter order will be high with narrow transition bands
- Filter order is independent of passband width
- Ripples have less significant than transition band

- ◆ Max. allowable deviation in attenuation in passband, $A_{\max}$:

$$A_{\max} = 20 \log_{10} \left(\frac{1 + \delta_c}{1 - \delta_c} \right) \approx 17.37 \delta_c \quad [dB]$$

- ◆ Min. attenuation in stopband, $A_{\min}$:

$$A_{\min} = 20 \log_{10} \left(\frac{1 + \delta_c}{\delta_s} \right) \quad [dB]$$



FIR filters Example (1)

Example

Synthesize a linear-phase FIR filter with $\omega_c T = 0.3\pi$, $\omega_s T = 0.6\pi$, $\delta_c = 0.02$, and $\delta_s = 0.0025$ (lowpass). Use McClellan-Parks-Rabiner's algorithm `remez.m`

```
% Example Linear-Phase, lowpass FIR
wT = [0.3 0.6]*pi;           % Band edges
bands = [1 0];               % Gain in the bands
deltas = [0.02 0.0025];      % Acceptable deviations
fsample = 2*pi;
[N, Be, D, W] = remezord(wT, bands, deltas, fsample);
N                               % Estimated filter order

h = remez(N, Be, D, W);
points = 16;
w = linspace(0, 180, 180*points+1); % wT axis
H = freqz(h,1, w*pi/180);      % Transfer function
Mag = 20*log10(abs(H));        % Magnitude response [dB]
Attenu = -Mag;                 % Attenuation [dB]
Phase = angle(H)*180/pi;       % Phase response [deg]
gd = grpdelay(h, 1, w*pi/180); % Group delay
```



FIR filters Example (2)

In this case we get from `remezord`

$$N = 13 \quad Be = [0 \ 0.3 \ 0.6 \ 1]^T \quad D = [1 \ 1 \ 0 \ 0]^T \quad W = [1 \ 8]^T$$

The filter order N that is obtained from `remezord` is only an estimation. It may therefore be necessary to try different filter orders in order to find a filter that satisfies the specification.

$$N = 13 \Rightarrow A_{min} \approx 51.7 \text{ dB and } A_{max} \approx 0.366 \text{ dB}$$

$$N = 14 \Rightarrow A_{min} \approx 50.9 \text{ dB and } A_{max} \approx 0.384 \text{ dB}$$

$$N = 15 \Rightarrow \text{neither acceptable}$$

$$N = 16 \Rightarrow A_{min} \approx 54.9 \text{ dB and } A_{max} \approx 0.26 \text{ dB.}$$

The symmetric impulse response is

$$h(0) = -0.003428964 = h(16)$$

$$h(1) = 0.002941935 = h(15)$$

$$h(2) = 0.021664496 = h(14)$$

$$h(3) = 0.015363454 = h(13)$$

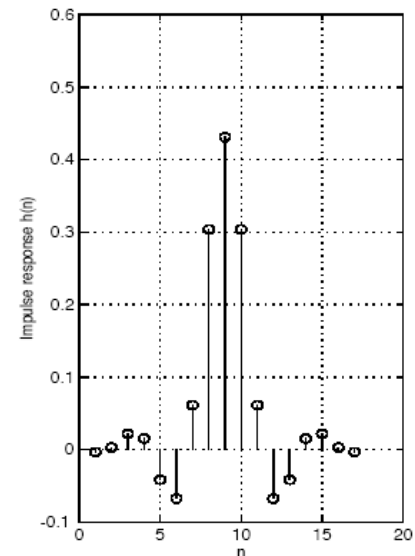
$$h(4) = -0.041601459 = h(12)$$

$$h(5) = -0.067766346 = h(11)$$

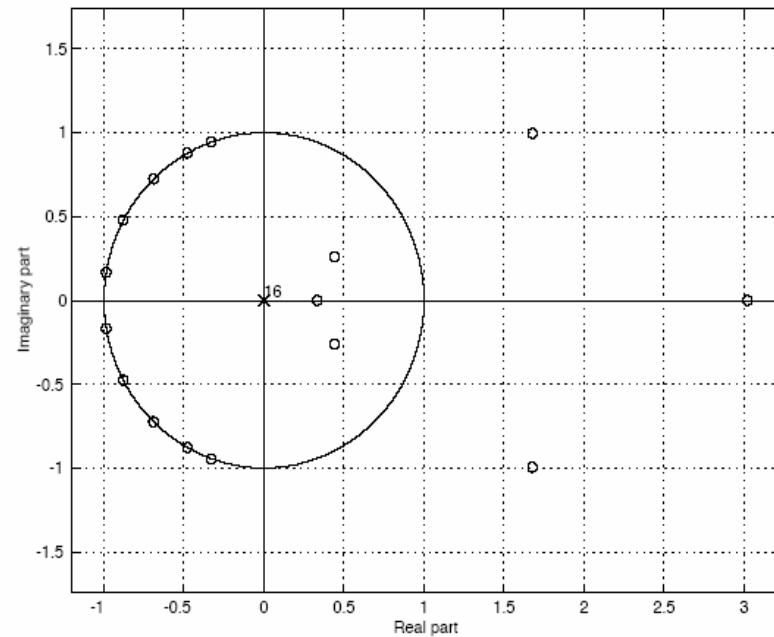
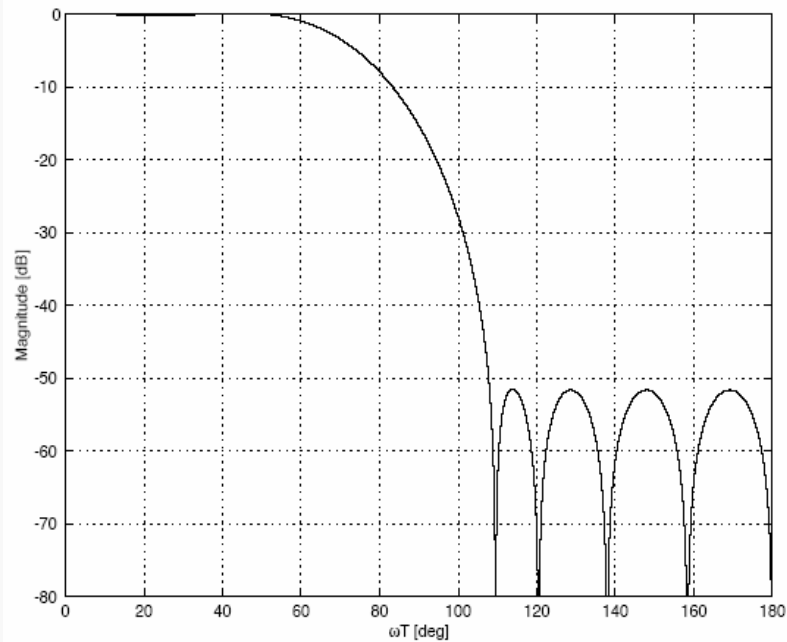
$$h(6) = 0.061181730 = h(10)$$

$$h(7) = 0.303702325 = h(9)$$

$$h(8) = 0.430966079$$



FIR filters Example (3)



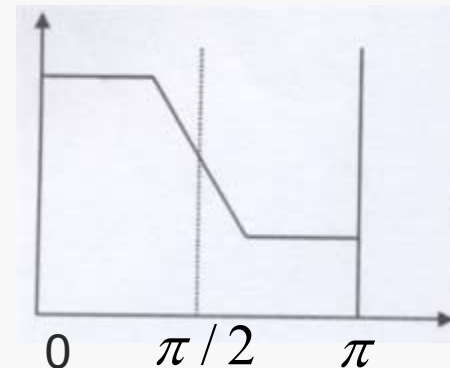
Half-Band FIR Filters (1)

- ◇ In many DSP Schemes, a large number of values in impulse response of certain types of FIR filters are zero
 - The required number of arithmetic operations can be reduced for such filters
- ◇ If the zero-phase function of an even-order lowpass, linear-phase FIR filter is antisymmetric with respect to $\pi/2$, i.e.,

$$H_R(e^{j\omega T}) = 1 - H_R(e^{j(\pi - \omega T)})$$

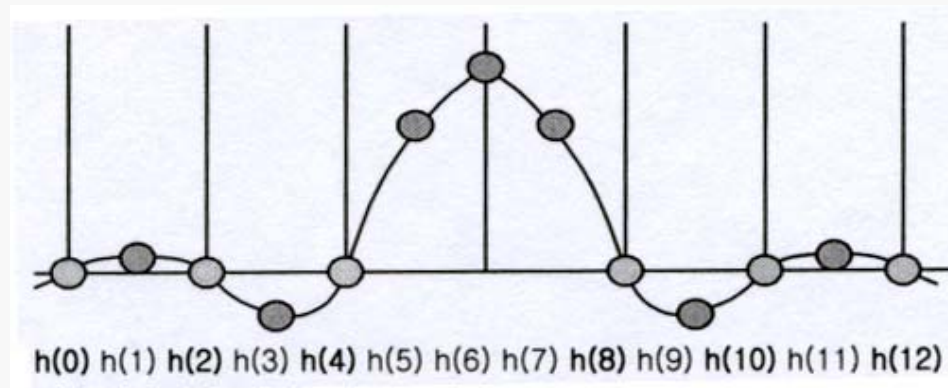
→ **Half-Band FIR filter**

(Bandwidth is about half of the whole freq. band)



Half-Band FIR Filters (2)

- ◇ Impulse response of half-band filter :



- ◇ Every other coefficient in the impulse response is zero except for the one in the center
- ◇ Large stopband attenuation results in small ripple in passband
- ◇ Required filter order is slightly higher than corresponding FIR filter

Design of Half-Band FIR filter

- ◆ Determine impulse response for half-band FIR filter that meets the following spec. :

■ $f_c = 400 \text{ KHz}$, $f_s = 600 \text{ KHz}$, $f_{\text{sample}} = 2 \text{ MHz}$, $A_{\text{min}} = 60 \text{ dB}$ ($\delta_2 = 0.001$)

● $A_{\text{max}} \approx 0.01737 \text{ dB}$

■ $N \approx 34.5 \Rightarrow 35$

- ◆ Impulse response

	Band 1	Band 2
Weighting	1.00000000	1.00000000
Deviation	0.00067458	0.00067458
Deviation in dB	0.01171863	-63.41935048

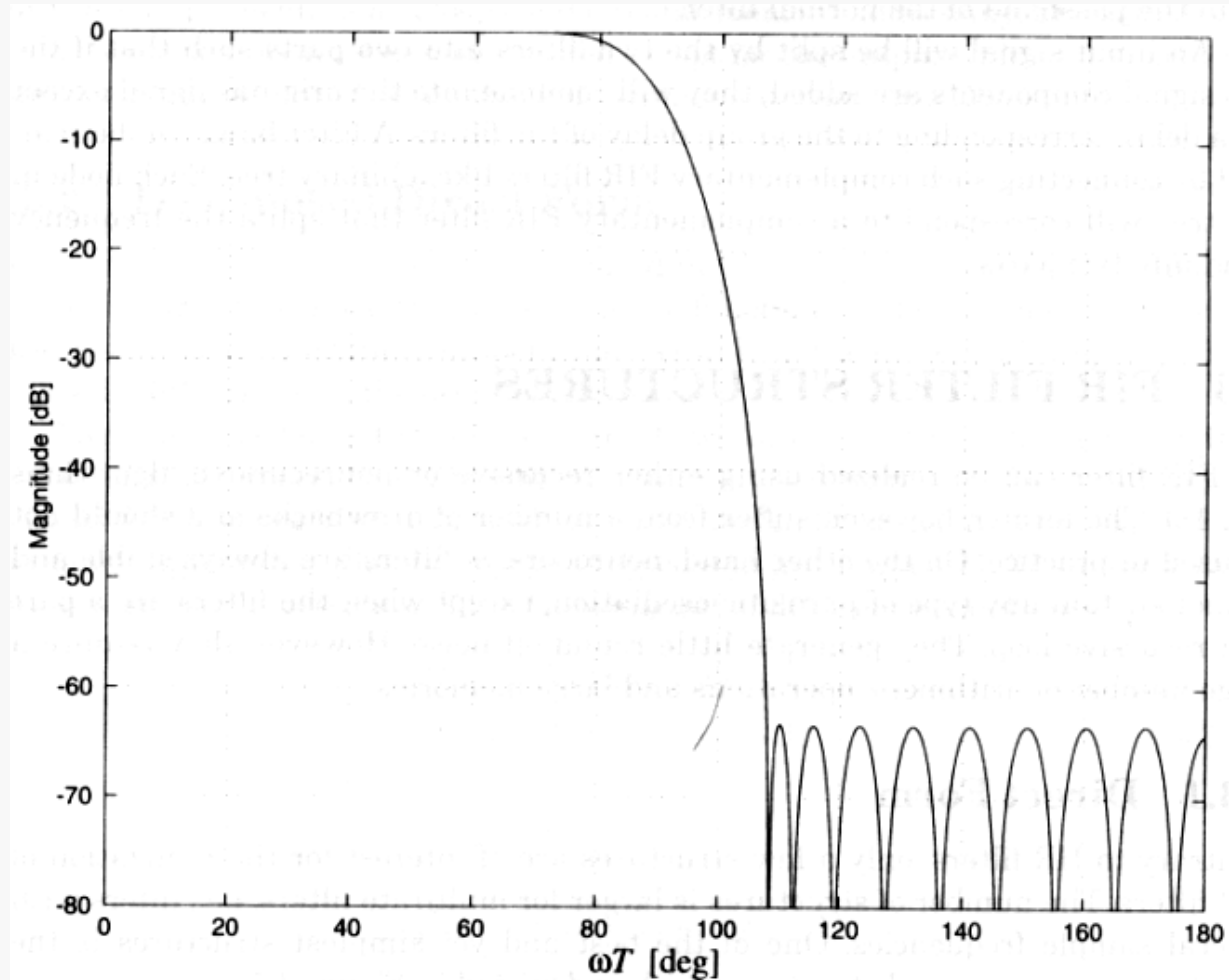
* * * * Impulse Response * * * *

$h(0) = 0.0011523147 = h(34)$	$h(9) = 0.0000205816 = h(25)$
$h(1) = 0.0000053778 = h(33)$	$h(10) = -0.0310922116 = h(24)$
$h(2) = -0.0027472085 = h(32)$	$h(11) = -0.0000213219 = h(23)$
$h(3) = -0.0000047933 = h(31)$	$h(12) = 0.0525993770 = h(22)$
$h(4) = 0.0057629677 = h(30)$	$h(13) = 0.0000279603 = h(21)$
$h(5) = 0.0000118490 = h(29)$	$h(14) = -0.0991307839 = h(20)$
$h(6) = -0.0107343126 = h(28)$	$h(15) = -0.0000262782 = h(19)$
$h(7) = -0.0000127702 = h(27)$	$h(16) = 0.3159246316 = h(18)$
$h(8) = 0.0185863941 = h(26)$	$h(17) = 0.5000310310$

* All odd-numbered coefficients are zero (small rounding errors)



Mag. response of H-B filter



FIR filter structures (1)

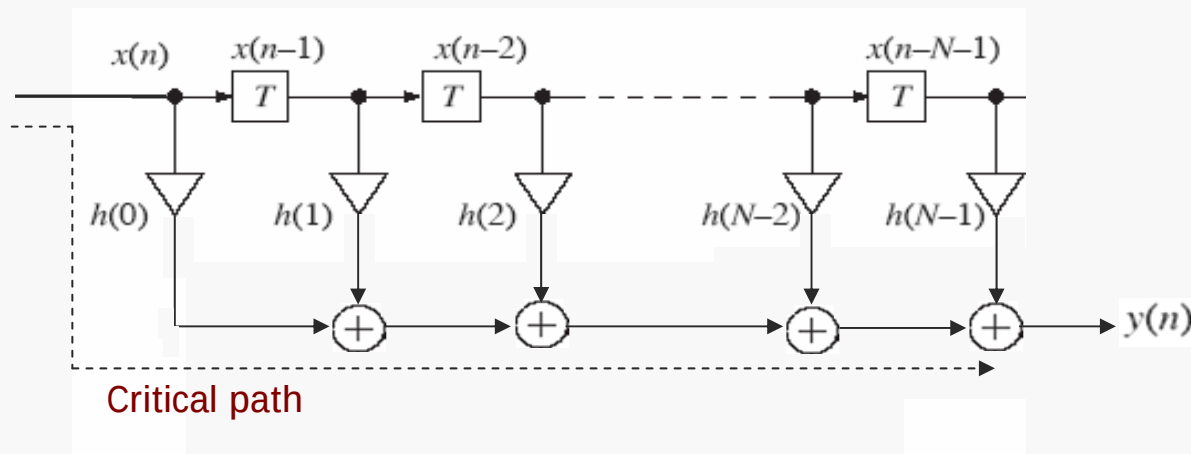
- ◇ Direct form (or transversal structure)
 - Suitable for system which is efficient in computing some-of-products
- ◇ Transposed direct form
 - Amount of required hardware can be significantly reduced for this type of structure where many multiplications are performed with the same input value
- ◇ Direct form linear-phase FIR structure
 - Advantage of linear-phase FIR filters is that the number of multiplications can be reduced by exploiting the symmetry (or antisymmetry) in impulse response



FIR filter structures (2)

◇ Direct form (or transversal structure) :

$$y(n) = \sum_{k=0}^{N-1} h(k) \cdot x(n-k)$$



◇ N multiplications and (N-1) additions

◇ Critical path : $T_C = T_M + (N-1) \cdot T_A$

where T_M : Multiplication time
 T_A : Addition time



FIR filter structures (3)

◇ Transposed Direct form :

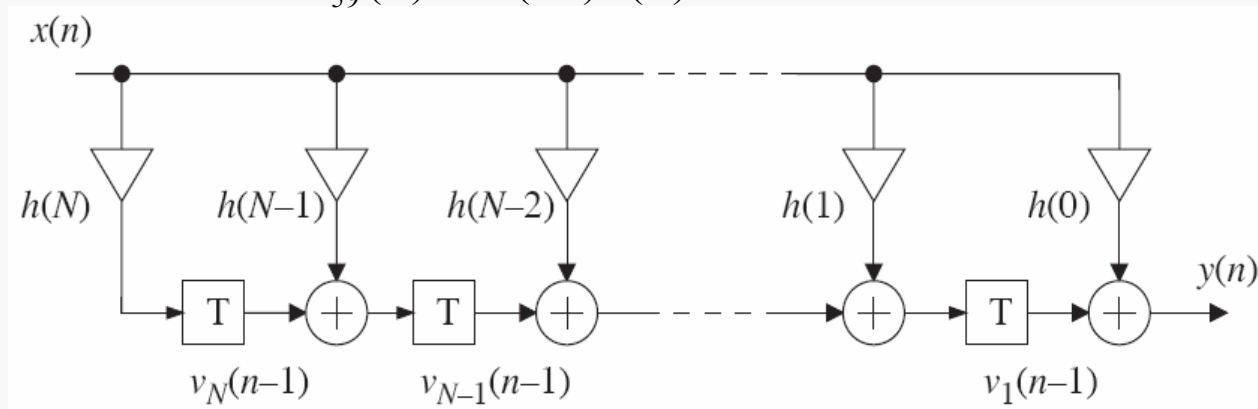
$$y(n) := h(0)x(n) + v_1(n-1)$$

$$v_1(n) := h(1)x(n) + v_2(n-1)$$

⋮

$$v_{58}(n) := h(58)x(n) + v_{59}(n-1)$$

$$v_{59}(n) := h(59)x(n)$$

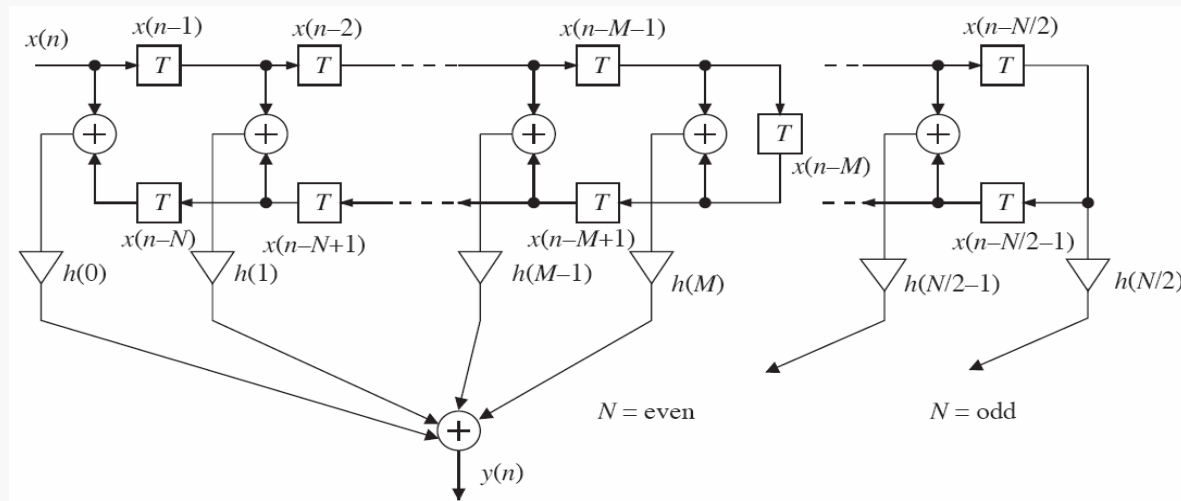


◇ Critical path : $T_C = T_M + T_A$

FIR filter structures (4)

◇ Direct form linear-phase FIR structure :

$$y(n) = \sum_{k=0}^{\frac{N}{2}} h(k)[x(n-k) \pm x(n-N+1+k)]$$



$$2M = N+1$$

- ◇ Multiplication is $(N+1)/2$ for $N=odd$ and $N/2$ for $N=even$
- ◇ Subtractors are used for antisymmetric case

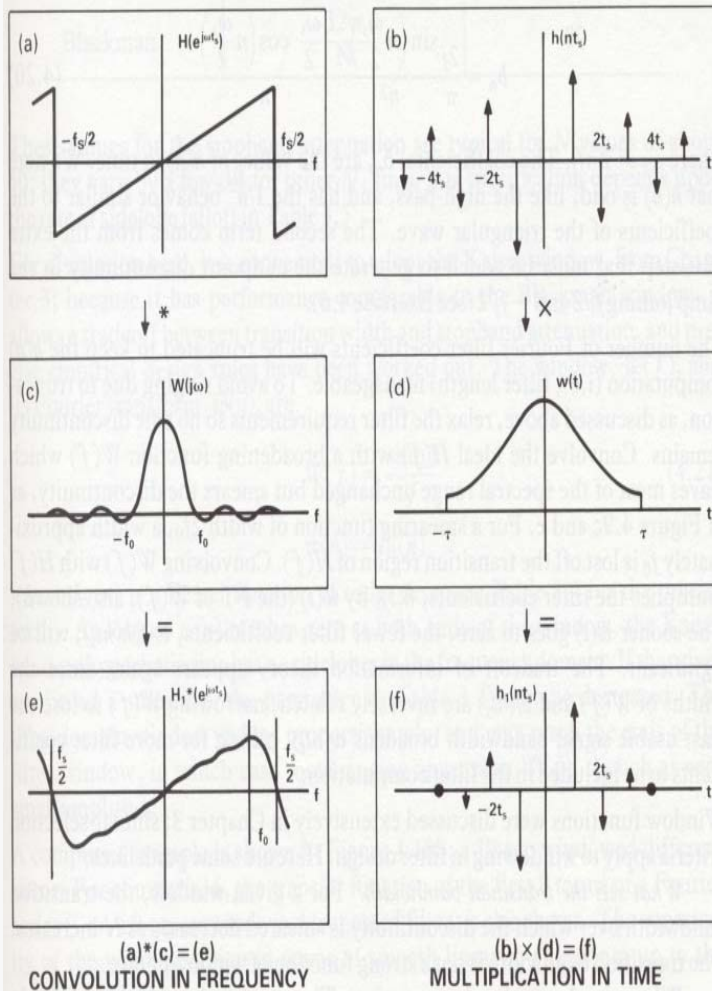


Facts underlying FIR filter design

- ◇ Filter coefficients in time are Fourier coefficients of the transfer function in frequency
- ◇ The sharper the cutoff demanded in $H(f)$, the more coefficients must be included in $h(n)$
- ◇ An ideally sharp cutoff of $H(f)$ is not desirable, since the step response in time will ring
- ◇ Truncation of the set, $h(n)$, at finite N results in Gibbs oscillations in $H(f)$
- ◇ Windowing softens the corners of $H(f)$, lowering the number of filter tap required



Windowing in FIR design



a) Ideal differentiator

$$H(\omega) = j\omega$$

b) Its Fourier coefficient in time

c) & d) window function

$$W(\omega) = w(t)$$

e) & f) windowed FIR results

Choosing Window Type

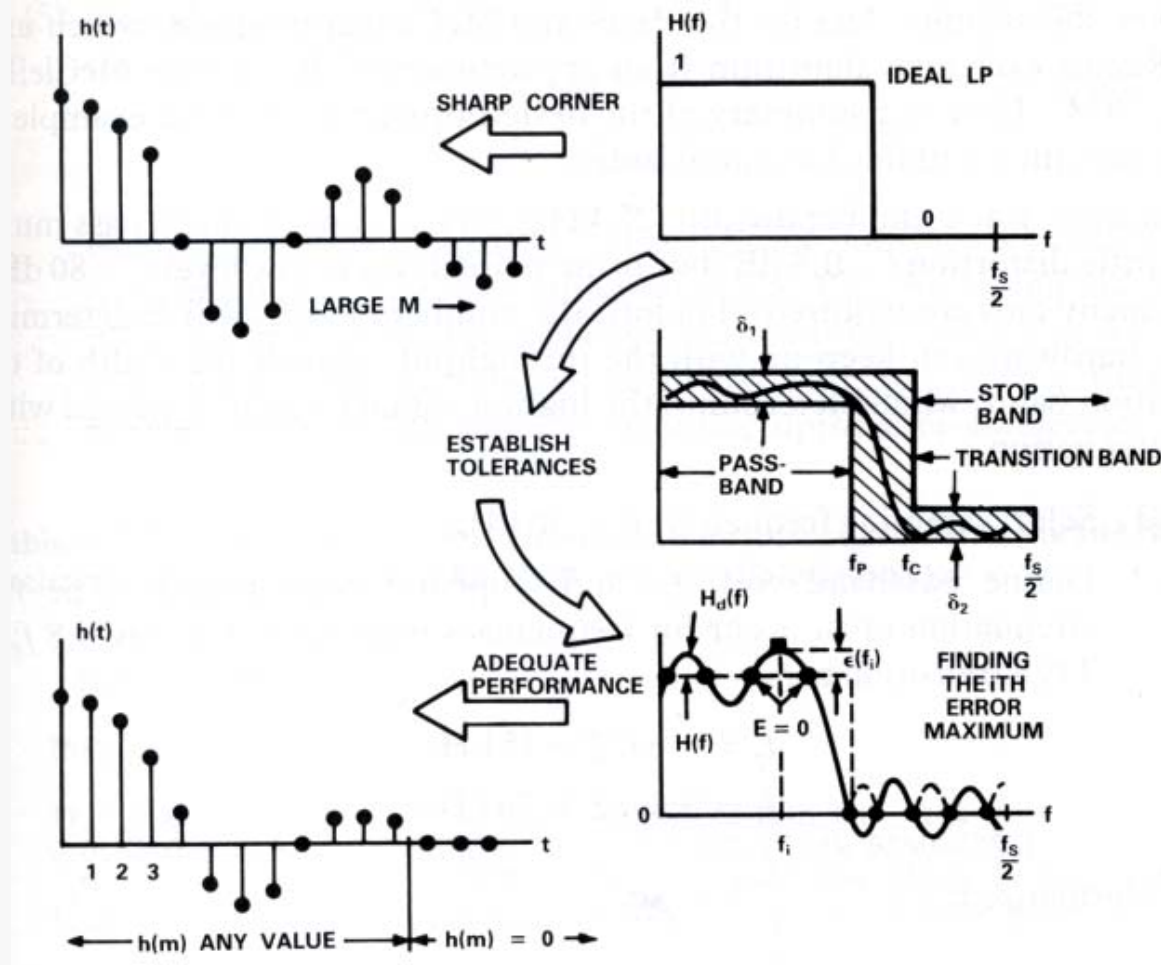
- ◇ Transition bandwidth is not a strong function of window choice
- ◇ Stopband attenuation is a strong function of window choice

Window	Transition Width	Max Stopband Attenuation (dB)
Rectangular	$\frac{4\pi}{N}$	20
Cosine Bell	$\frac{8\pi}{N}$	50
Hamming	$\frac{8\pi}{N}$	45
Blackman	$\frac{12\pi}{N}$	80

* Max stopband attenuation is for $N \approx 50$



CAD optimization for FIR filter design



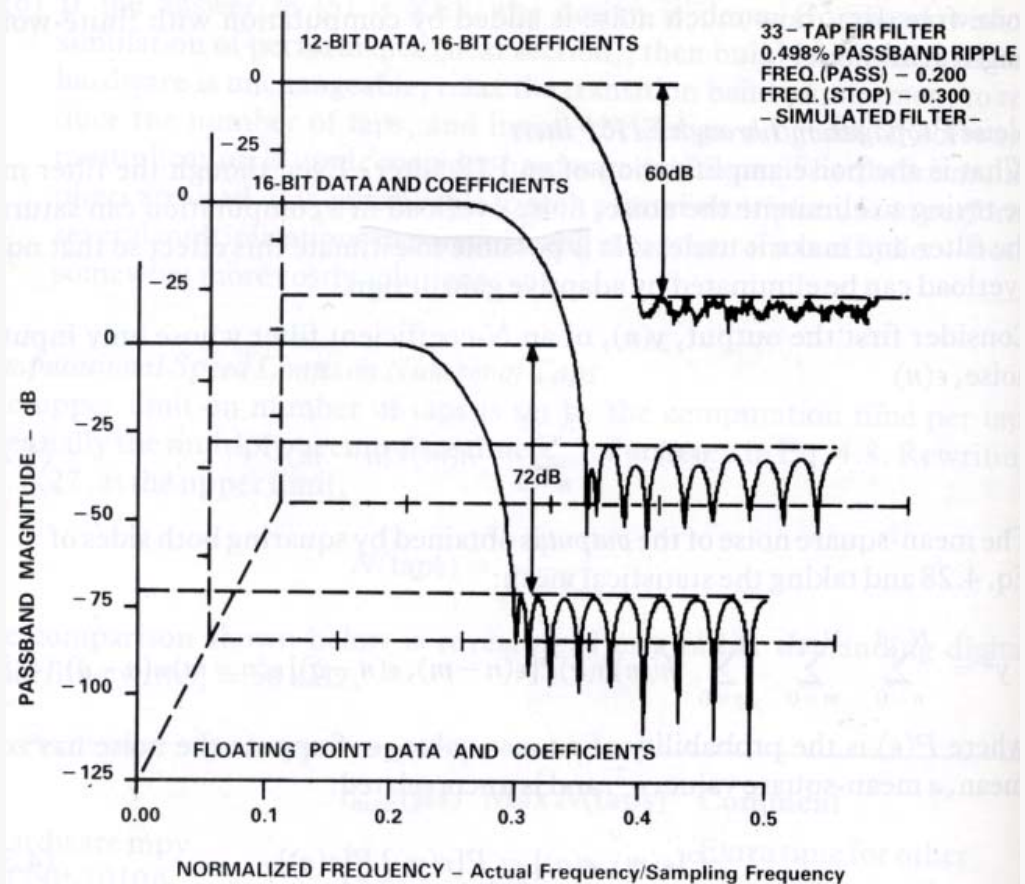
Fixed-point Implementation

- ◇ All data values and coefficients must be represented with a finite word length
- ◇ Word length is a crucial parameter for
 - Filter performance
 - Power consumption
 - Area (Size)
 - Clock speed
- ◇ Fixed-point quantization errors
 - Overflow of number range in output signal
 - Rounding errors of products (or truncation)
 - Deviation from the ideal frequency response for filter



Effects of finite word length

Data	Coeff.	Stopband Attenuation
float	float	72 dB
16bit	16bit	70 dB
12bit	16bit	60 dB

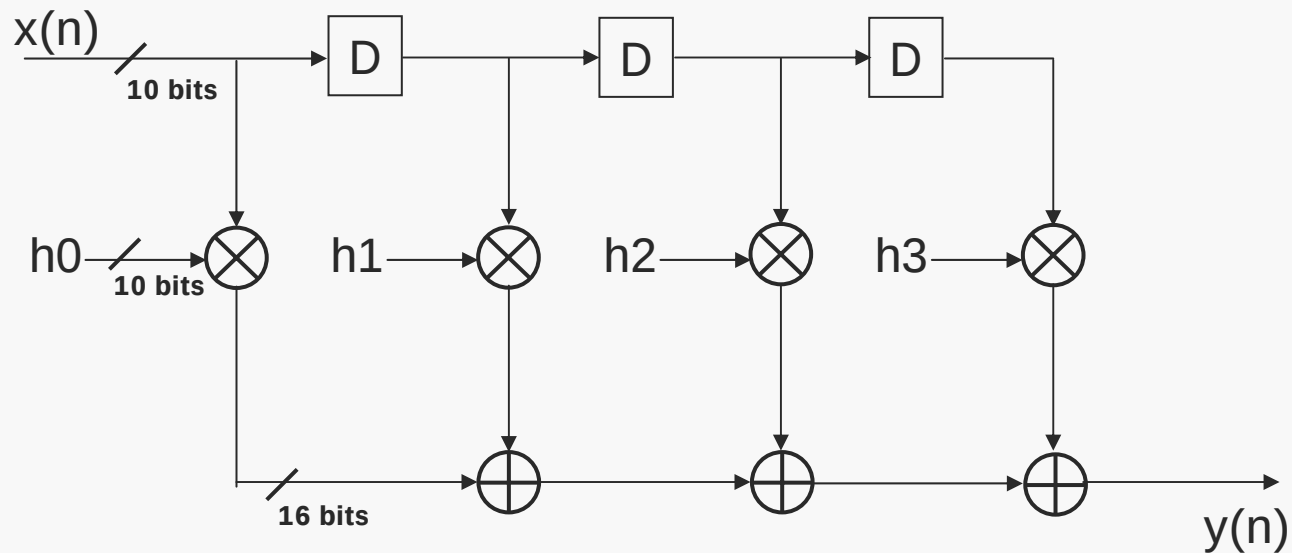


H/W Implementation Schemes of FIR

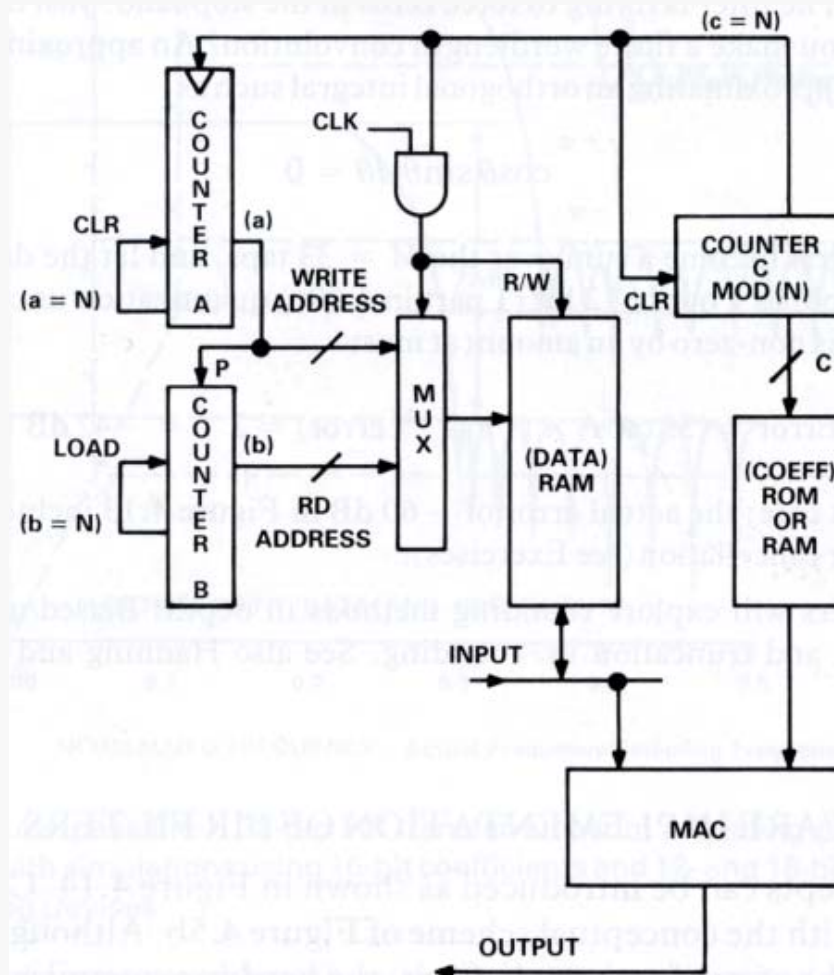
- ◇ Direct implementation of FIR structure
- ◇ General-purpose implementation
- ◇ LUT (Look-Up Table)-based implementation



Direct Implementation of FIR structure



General-purpose Implementation (1)



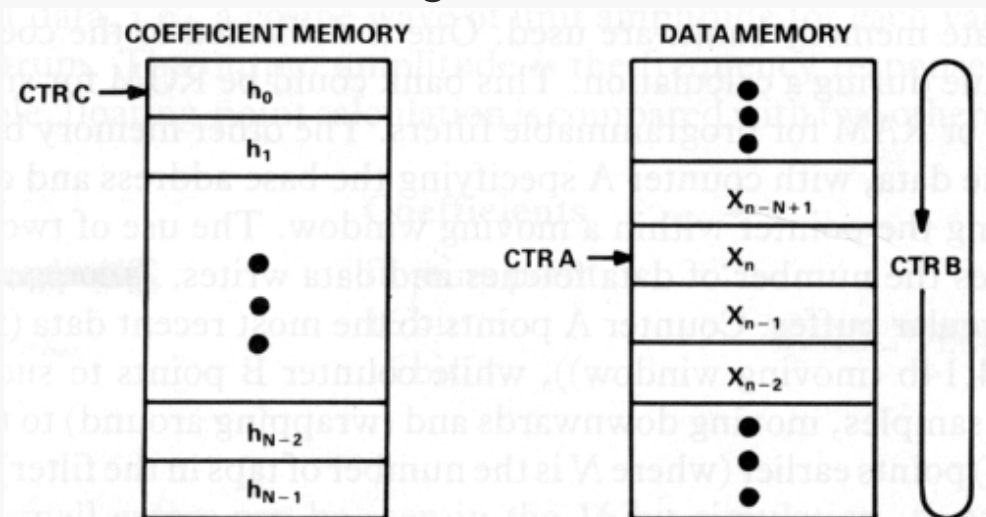
General-purpose Implementation (2)

◇ Coefficient memory :

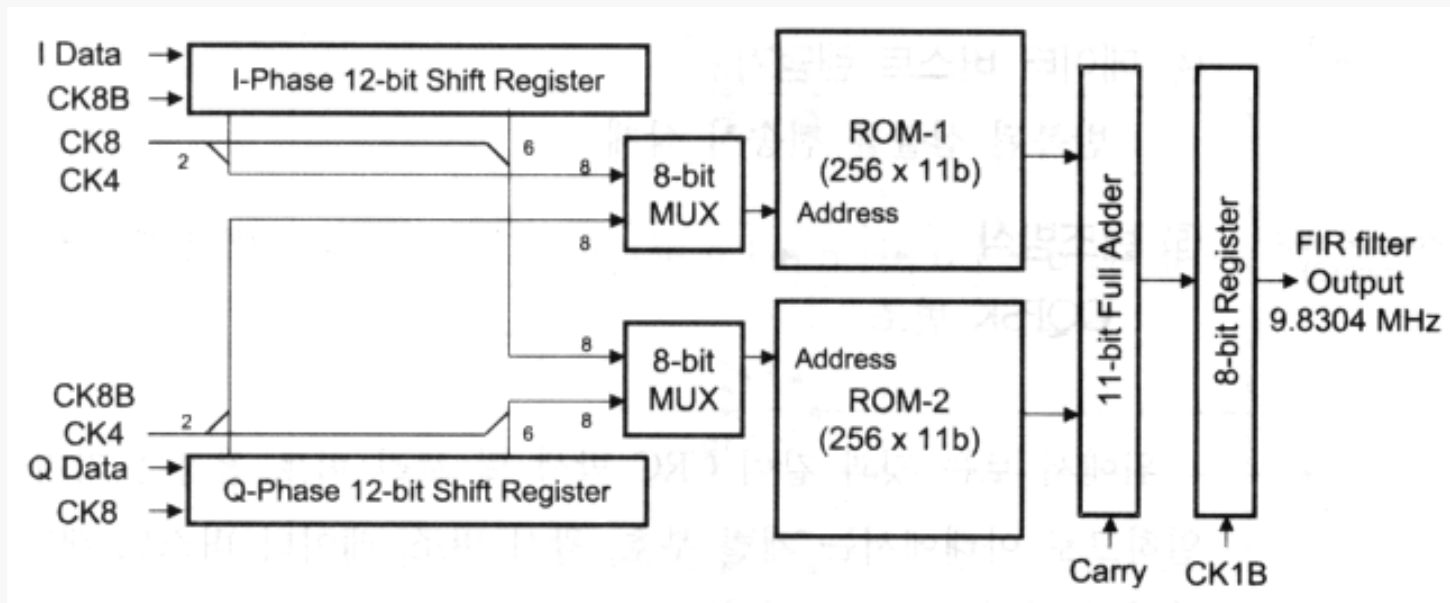
- ROM for specific filter or RAM for programmable filters

◇ Data memory :

- Circular buffer
- Counter A specifies base address, Counter B increments the pointer within a moving window



LUT-based Implementation



Comparison of Impl. Schemes

◇ Comparison of FIR Filter Implementation Schemes

	Direct	General-purpose	LUT-based
Complexity (Gate Count)	High	Low	Medium
Clock* Frequency	Low	High	Low
Latency (Proc. Delay)	Medium	High	Low

- Both direct and LUT-based implementation schemes need the same clock frequency.



IIR Filters

IIR filters

- ◆ IIR filter response of order N :

$$y(n) = \sum_{k=0}^M a(k) \cdot x(n-k) + \sum_{k=1}^N b(k) \cdot y(n-k)$$

- ◆ Transfer function of a IIR filter :

$$H(z) = \frac{\sum_{k=0}^M a_k z^{-k}}{1 - \sum_{k=1}^N b_k z^{-k}}$$

- ◆ Poles can be placed anywhere within unit circle of Z-plane and they can be used to improve the frequency selectivity
- ◆ Required filter order is much lower than FIR filter
- ◆ Often use cascade two IIR filters to get a linear phase response in the passband
- ◆ It may be efficient to use a combination of FIR & IIR



Specification of IIR filters

- ◇ Typical attenuation spec. with passband & stopband frequencies
 - Passband frequency : f_c
 - Stopband frequency : f_s
 - Ripple in the passband : ε_p or $A_{\max} = 10\log_{10}(1 + \varepsilon_p^2)$
 - Attenuation in stopband : ε_s or $A_{\min} = 10\log_{10}(1 + \varepsilon_s^2)$
- ◇ In some applications, phase & group delay requirements must be met
 - Group delay variation within passband is typically specified to be within certain limits for acceptable signal distortion



Analog filter approximations

- ◇ Classical analog LPF approximations can be used for IIR filter design

Filter Type	Magnitude	Group delay
Butterworth	- maximally flat at origin - monotonically decreasing in PB & SB	- variation is comparatively small in PB - OGD is large
Chebyshev I	- equal ripple in PB - monotonically decreasing in SB	- variation is worse than Butterworth (BW) - OGD is smaller than BW
Chebyshev II	- maximally flat at origin - monotonically decreasing in PB - equal ripple in SB	- variation is similar to BW - much smaller OGD
Cauer (Elliptic)	- equal ripple in PB & SB	- variation is large - OGD is the smallest
Bessel	- maximally flat at origin - monotonically decreasing in PB - SB attenuation is poor	

* **PB** : Passband, **SB** : Stopband, **OGD** : Overall Group Delay



Design methods of IIR filters (1)

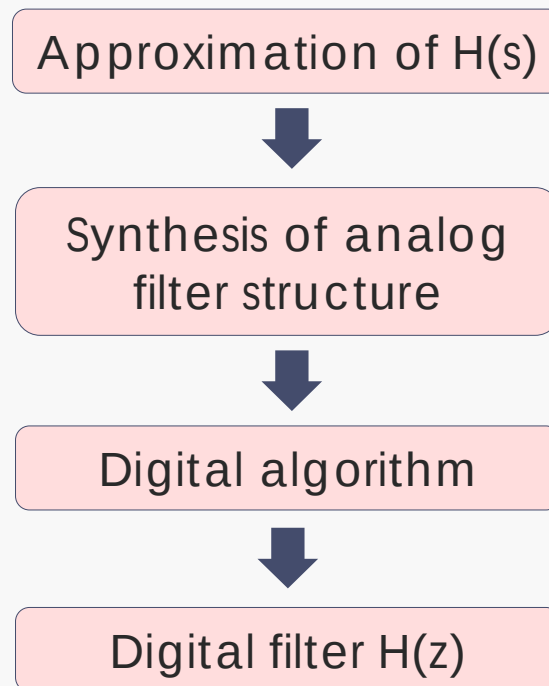
- ◇ 1. Direct design in Z-plane
 - Can be designed by directly placing the poles & zeros in Z-plane such that the freq. response satisfies the requirement
 - Approximation methods used in analog filter design can be used
- ◇ 2. Mapping of analog transfer functions
 - Magnitude spec. of digital filter is mapped to an equiv. spec. for analog filter using bilinear transformation method
 - Approximation of analog transfer function $H(s)$ is solved
 - Mapping of poles & zeros from S-plane to Z-plane for digital filter & finding a digital transfer function $H(z)$

* Digital LPF $\rightarrow$ BPF by transformation $z \rightarrow -z^2$



Design methods of IIR filters (2)

◇ 3. Mapping of analog filter structures



Biquad : Building block of IIR filter

- ◇ 2nd-order Recursive filter with difference equation for $n=2$

Difference equation for $n = 2$

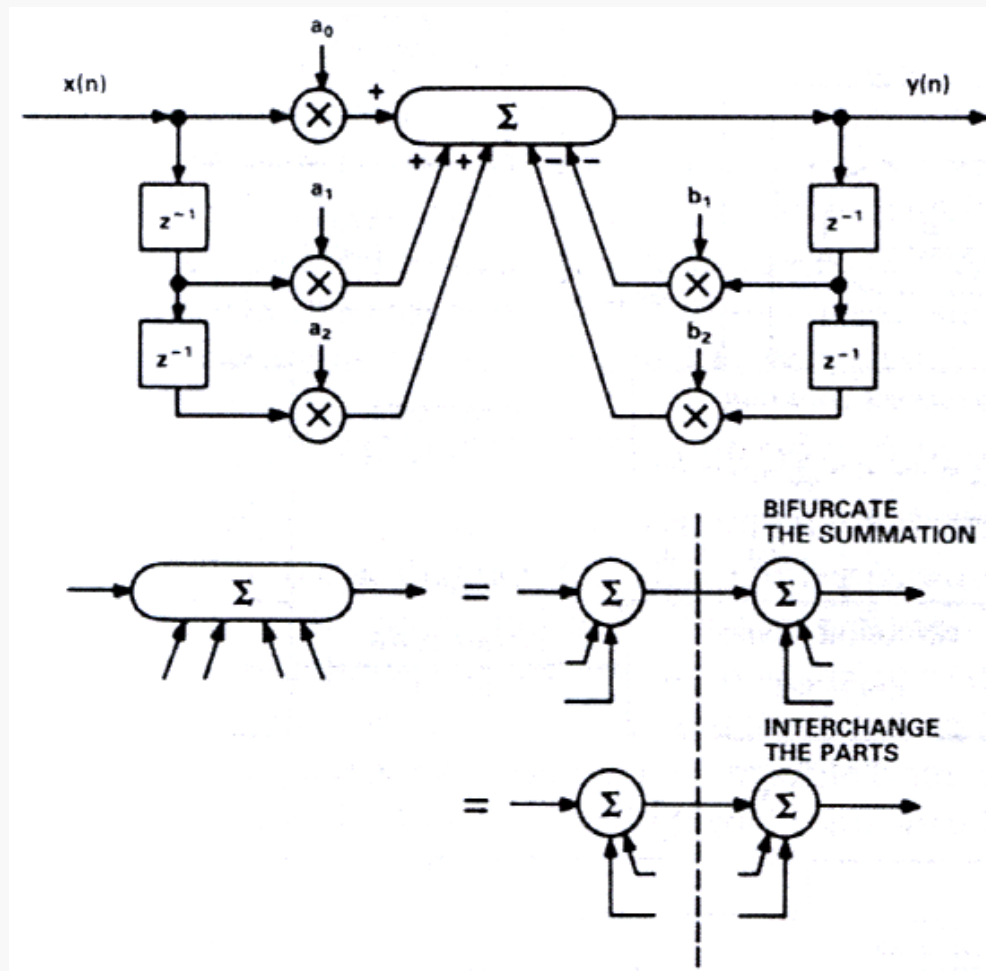
$$\begin{aligned} y(n) &= \sum_{k=0}^2 a_k x(n-k) + \sum_{k=1}^2 b_k y(n-k) \\ &= a_0 x(n) + a_1 x(n-1) + a_2 x(n-2) + b_1 y(n-1) + b_2 y(n-2) \end{aligned}$$

$$Y(z) = \sum_{k=0}^2 a_k z^{-k} X(z) + \sum_{k=1}^2 b_k z^{-k} Y(z)$$

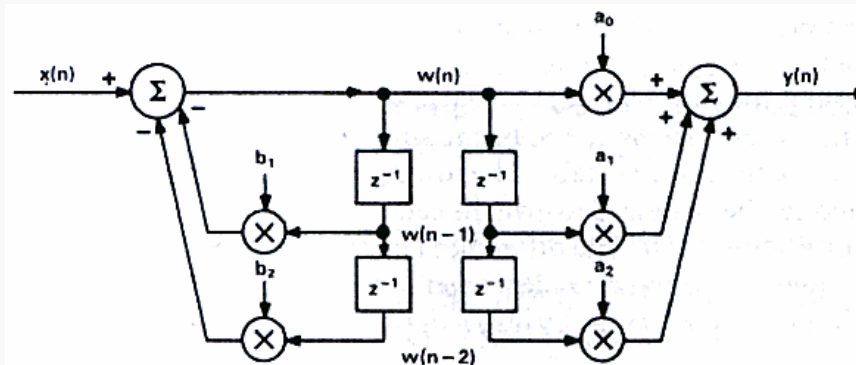
$$\begin{aligned} H(z) &= Y(z) / X(z) \\ &= \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 - b_1 z^{-1} + b_2 z^{-2}} \end{aligned}$$



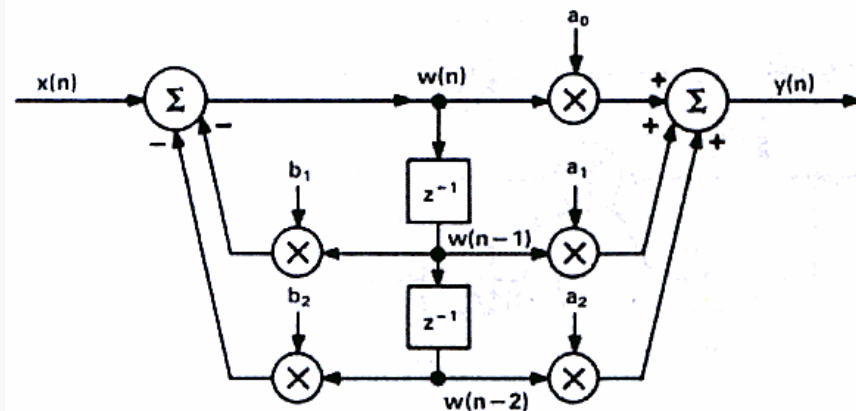
Biquad filter, direct form



Biquad filter, canonical form



a. Reverse cascade of the filter in Figure 4.16.



b. Equivalent operations coalesced.

Relationship between coefficients and pole & zero locations

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 - b_1 z^{-1} + b_2 z^{-2}}$$
$$= G \frac{(z - r_0 e^{-j\theta_0})(z - r_0 e^{+j\theta_0})}{(z - r_p e^{-j\theta_p})(z - r_p e^{+j\theta_p})}$$

$$\Rightarrow \begin{aligned} a_0 &= 1 & a_1 &= -2r_0 \cos \theta_0 & a_2 &= +r_0^2 \\ b_0 &= 1 & b_1 &= +2r_p \cos \theta_p & b_2 &= -r_p^2 \end{aligned}$$

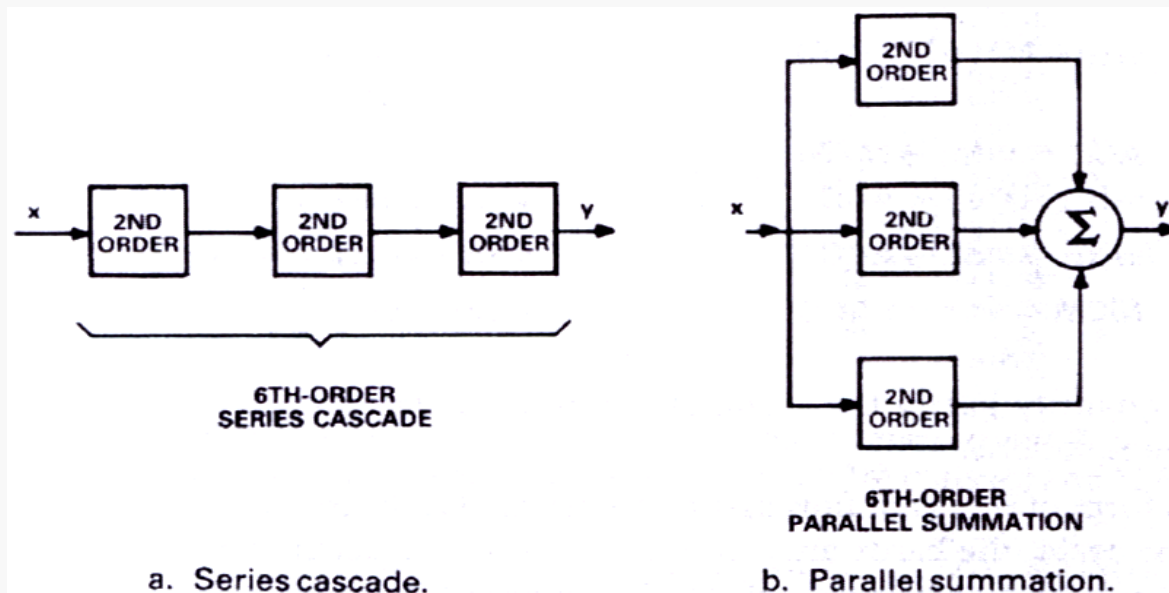


Design of Higher-order IIR filter

- ◇ As a series or parallel cascade of biquad modules

$$H(z) = \frac{\sum_{k=0}^M a_k z^{-k}}{\sum_{r=0}^N b_r z^{-r}} = B_1(z) \times B_2(z) \times B_3(z) \times \dots$$

$$H(z) = \sum_{r=1}^{\frac{N}{2}} \frac{a_{0r'} + a_{1r'} z^{-1}}{1 - b_{1r'} z^{-1} - b_{2r'} z^{-2}} + \sum_{i=0}^{M-N} c_i z^{-i}$$





Multirate Signal Processing

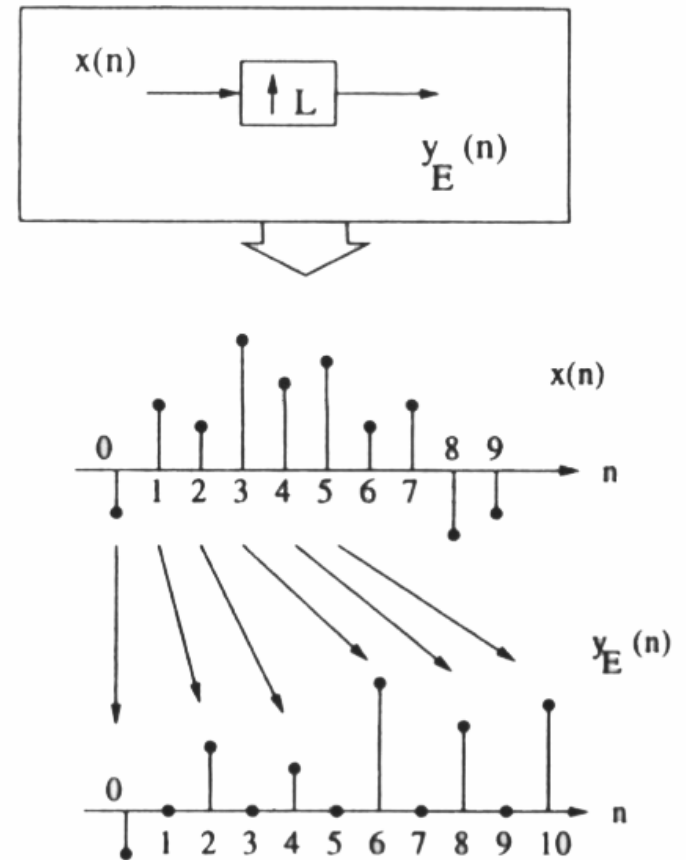
Multirate Signal Processing

- ◆ Transfer signals between two systems that operate with different sample frequencies
- ◆ Implemented System functions more efficiently by using several sample rates (Ex. narrow-band filters)
- ◆ Sample rate converters for changing the sample rate without affecting the information contained in the signal
- ◆ Converters for increasing and decreasing the sample frequency are usually called **interpolators** and **decimators**, respectively
- ◆ Potential advantages of multirate signal processing are reduced computational work load, lower filter order, lower coefficient sensitivity and noise, and less stringent memory requirements
- ◆ Disadvantages are more complex design, aliasing and imaging errors and a more complex algorithm
- ◆ Multirate techniques are used today in many digital signal processing systems



Interpolation (1)

- ◆ The process of increasing the sampling rate is called *interpolation*
- ◆ The aim is to get a new sequence corresponding to a higher sampling frequency, but with the same informational content, i.e., with the same spectrum as the underlying analog signal
- ◆ The interpolation process is essentially a two-stage process



Interpolation (2)

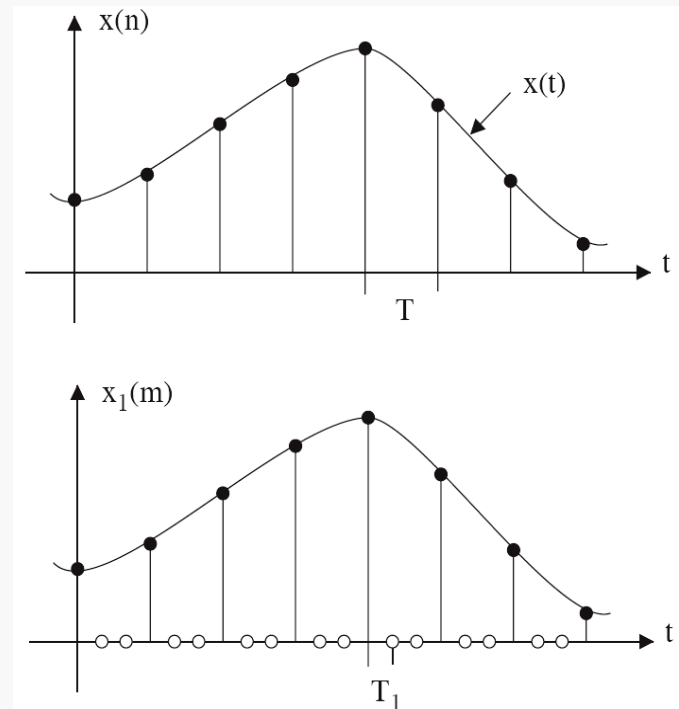
◇ First step

- An intermediate sequence $x_1(m)$ is generated from the original sequence $x(n)$ by inserting zero-valued samples between the original sequence values to obtain the desired sampling rate

$$x_1(m) = \begin{cases} x\left(\frac{m}{L}\right) & \text{if } m = 0, \pm L, \pm 2L, \dots \\ 0 & \text{otherwise} \end{cases}$$

- The sample period for the new sequence is

$$T_1 = T/L$$



Interpolation (3)

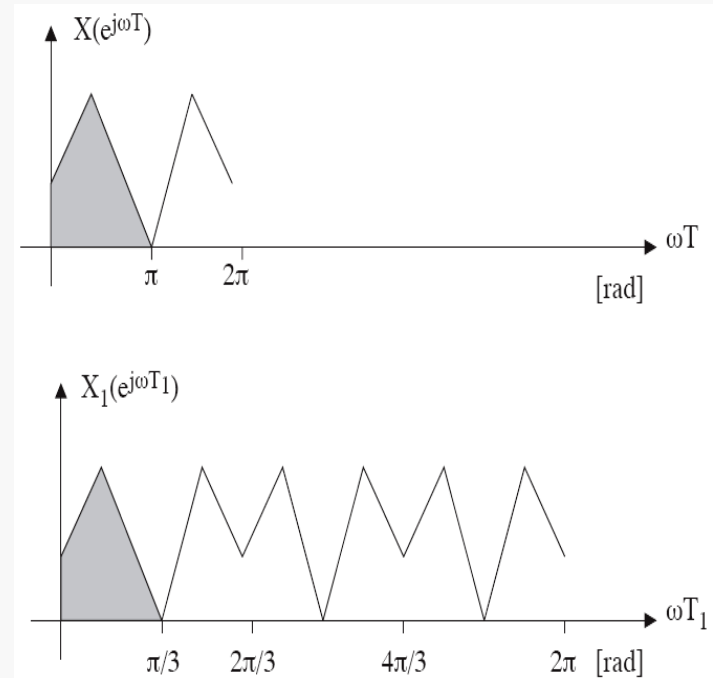
- The Fourier transform of $x_1(m)$ is

$$X_1(e^{j\omega T_1}) = \sum_{m=-\infty}^{\infty} x_1(m) e^{-j\omega m T_1} = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n L T_1} = X(e^{j\omega T})$$

- The original sequence $x(n)$ and the corresponding interleaved sequence $x_1(m)$ that has a three times higher sampling rate
- The spectrum of the sequence $x_1(m)$, contains not only the baseband

$$-\frac{\pi}{T_1} < \omega < \frac{\pi}{T_1}$$

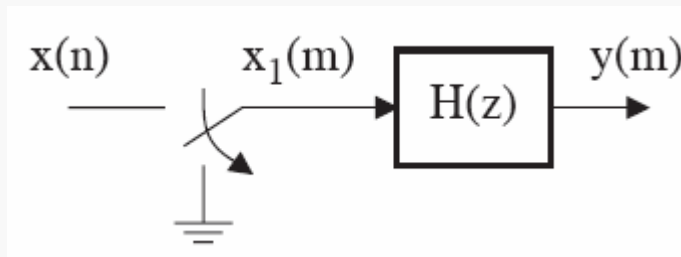
of the original signal, but also repeated images of the baseband



Interpolation (4)

◇ Second step

- An (ideal) lowpass filters are then used to remove the unwanted images
- Obviously, it is not necessary to perform arithmetic operations involving the zeros in the input sequence $x_1(m)$

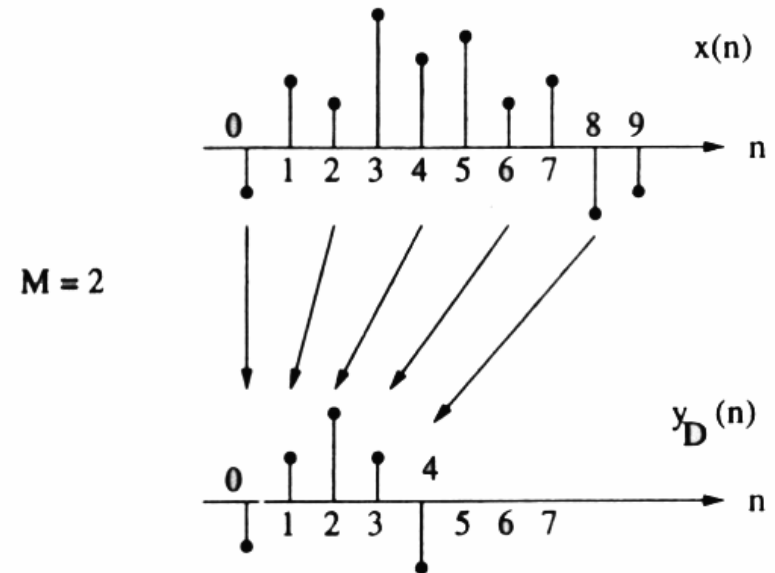
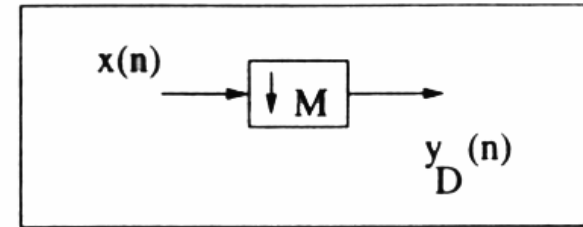


- Various schemes have been proposed to exploit this fact in order to reduce the computational work load

* Interpolation can be performed efficiently using FIR or IIR filter

Decimation (1)

- ◇ The process of reducing the sampling frequency is called decimation
- ◇ The sample frequency can be reduced by simply dropping every other sample



Decimation (2)

- ◆ Decimation (or down-sampler) process:

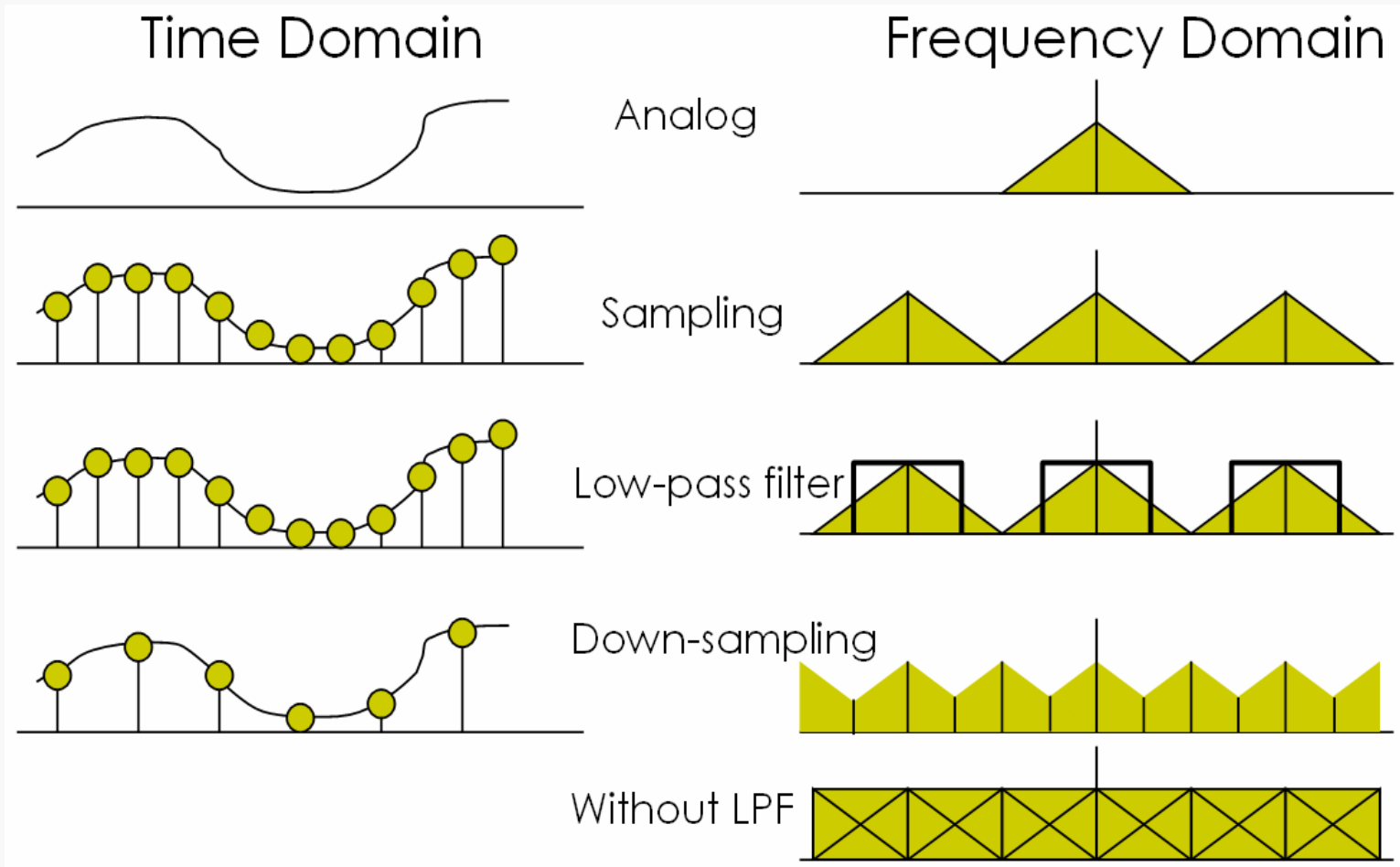
$$y_D(n) = x(M_n) \quad \text{where } M: \text{Positive integer}$$

- ◆ The output rate is M times slower than that of the input sequence
- ◆ Fourier transform of decimated signal consists of a sum of shifted replicas of F.T of original signal

$$X_D(e^{j\omega T}) = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\omega T - 2\pi k)/M})$$



Decimation (3)



Decimation (4)

* Example : x2 Decimation

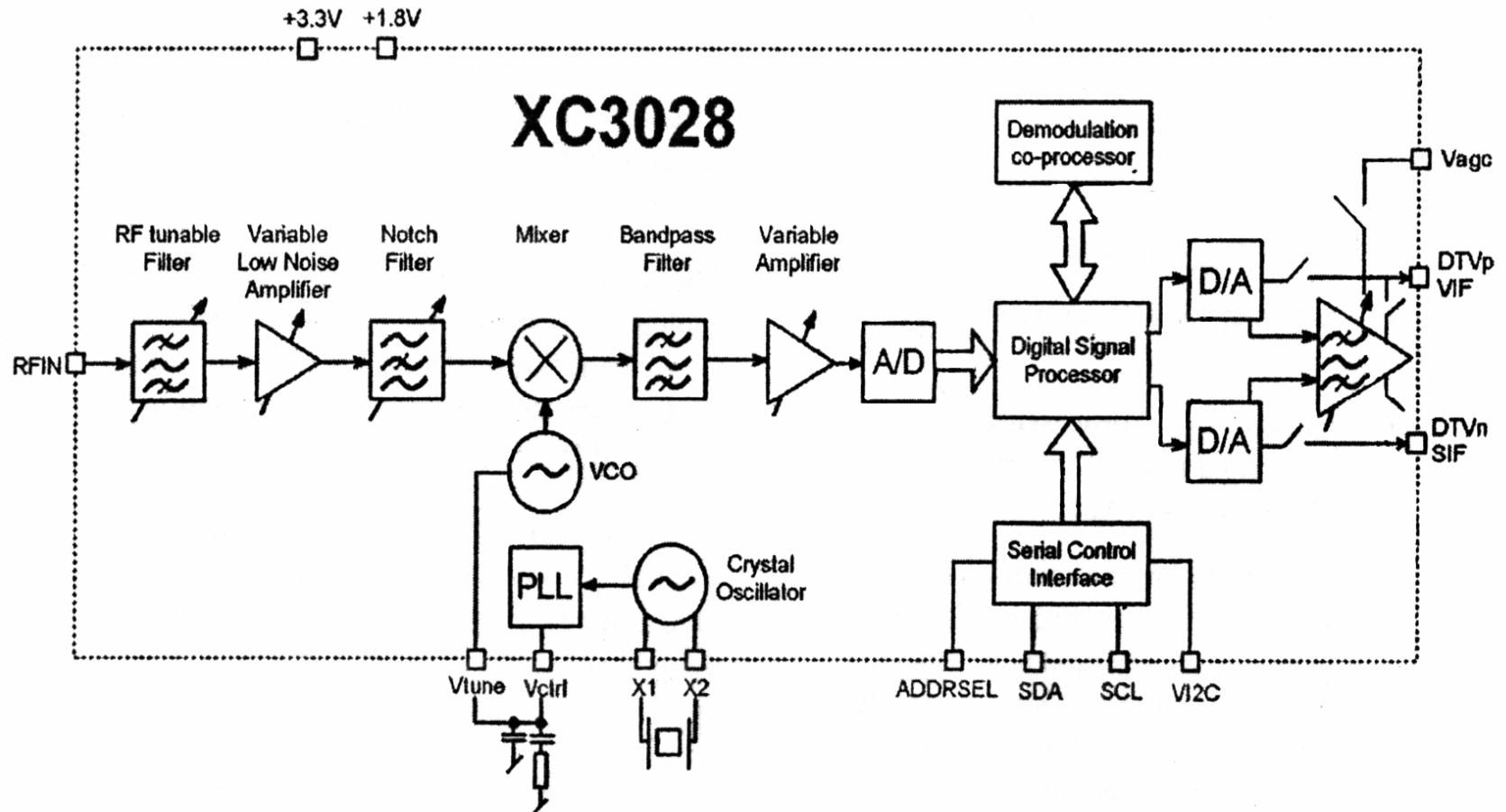
- ◇ Original sampling frequency 50KHz
- ◇ Input = 5KHz + 10KHz + 17KHz
- ◇ X2 down-sampling by decimation
 - Now the sampling frequency becomes 25KHz
- ◇ Observe two different outputs
 - Decimation without LPF
 - Decimation after LPF
 - 37tap FIR filter
 - Passband = 11KHz, stopband = 14KHz
 - Attenuation = 60dB



Two thick, curved lines are positioned around the text. A light purple line starts on the left and curves upwards towards the text. A darker purple line starts above the text and curves downwards towards the right.

Design Example of Digital SAW Filter

Block Diagram of Digital Tuner



Spec. of DTV signal

- ◇ Two modes of DTV operation
 - DTV6 : 6 MHz channel bandwidth
 - DTV8 : 8 MHz channel bandwidth
- ◇ DTV signal specification

Item		Specification
Input Signal		10 bit ADC(@ 25 MHz??)
Nominal center freq.	VSB/QAM	44 MHz
	OFDM	36 MHz
Bandwidth	VSB/QAM	6 MHz
	OFDM	6/8 MHz
In-band ripple		0.2 dB max
Max. In-band SNR for DTV8		50 dB
Adjacent Channel Rejection		70 dB
Phase Noise		-85 dBc/Hz (@10 kHz)



Design of Digital SAW filter

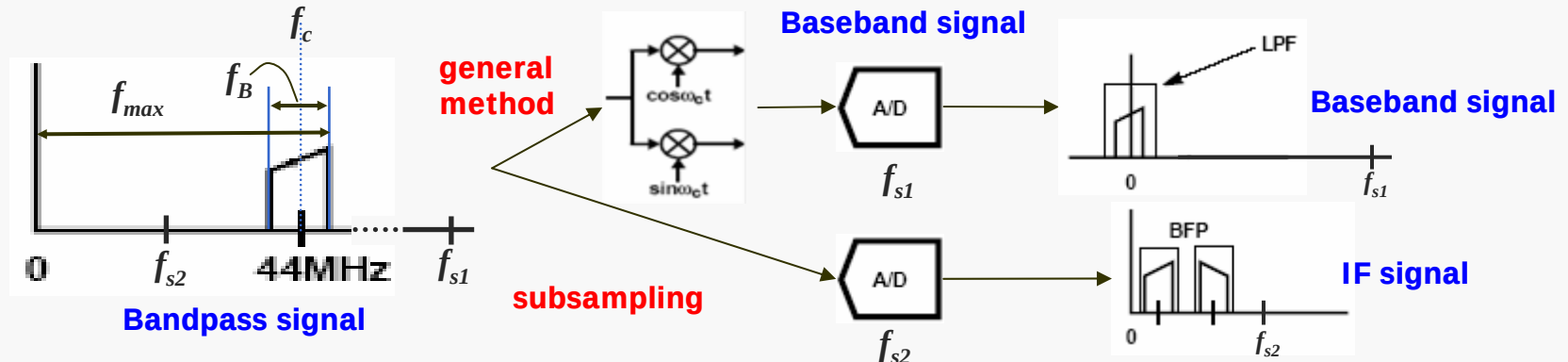
◇ Center Frequency Issue

■ General Method

- Downconversion bandpass signal to baseband signal
- AD conversion (digital baseband signal) : $2f_{max} < f_{s1}$
- Lowpass filtering

■ Subsampling

- AD conversion : $f_B < f_{s2} < f_c$
- Bandpass filtering



Subsampling Case

◆ Ideal sampling (impulse-train sampling)

$$x_p(t) = x(t)p(t) \quad (\text{where } p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s))$$

$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$$

$$X_p(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$$

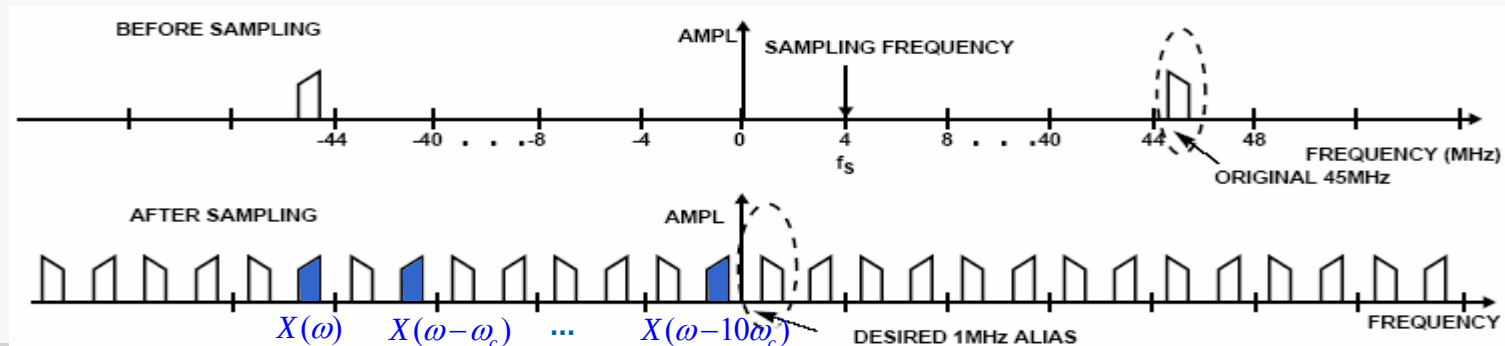
replica of original signal

◆ Subsampling example

$$X(\omega - \omega_s)$$

■ Signal center freq. : 45 MHz, sampling freq. : 4 MHz

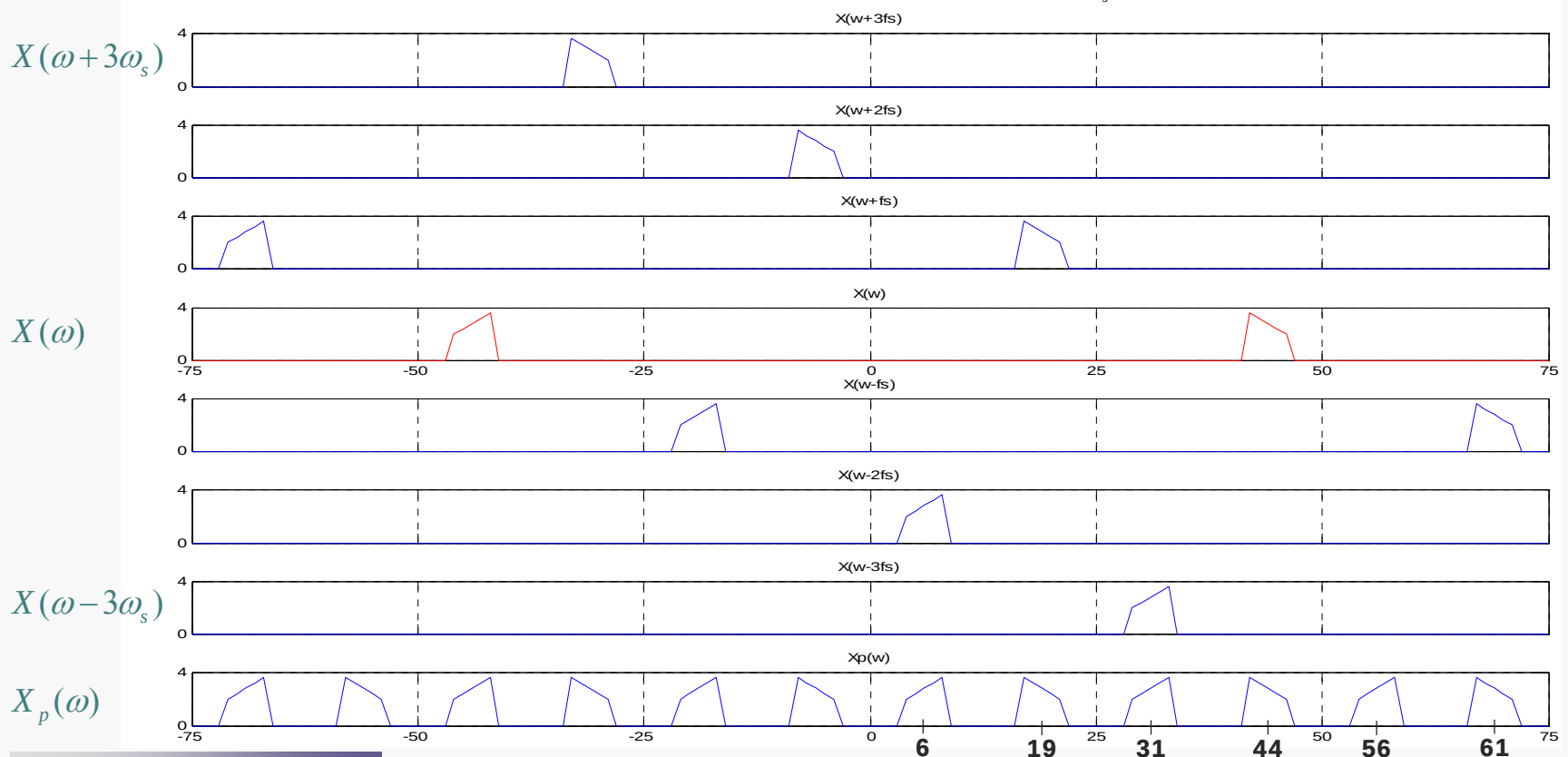
■ Converted center freq. : 1 MHz



Subsampling DTV signal (1)

◇ Case 1 : VSB/QAM (44MHz $\pm$ 3MHz)

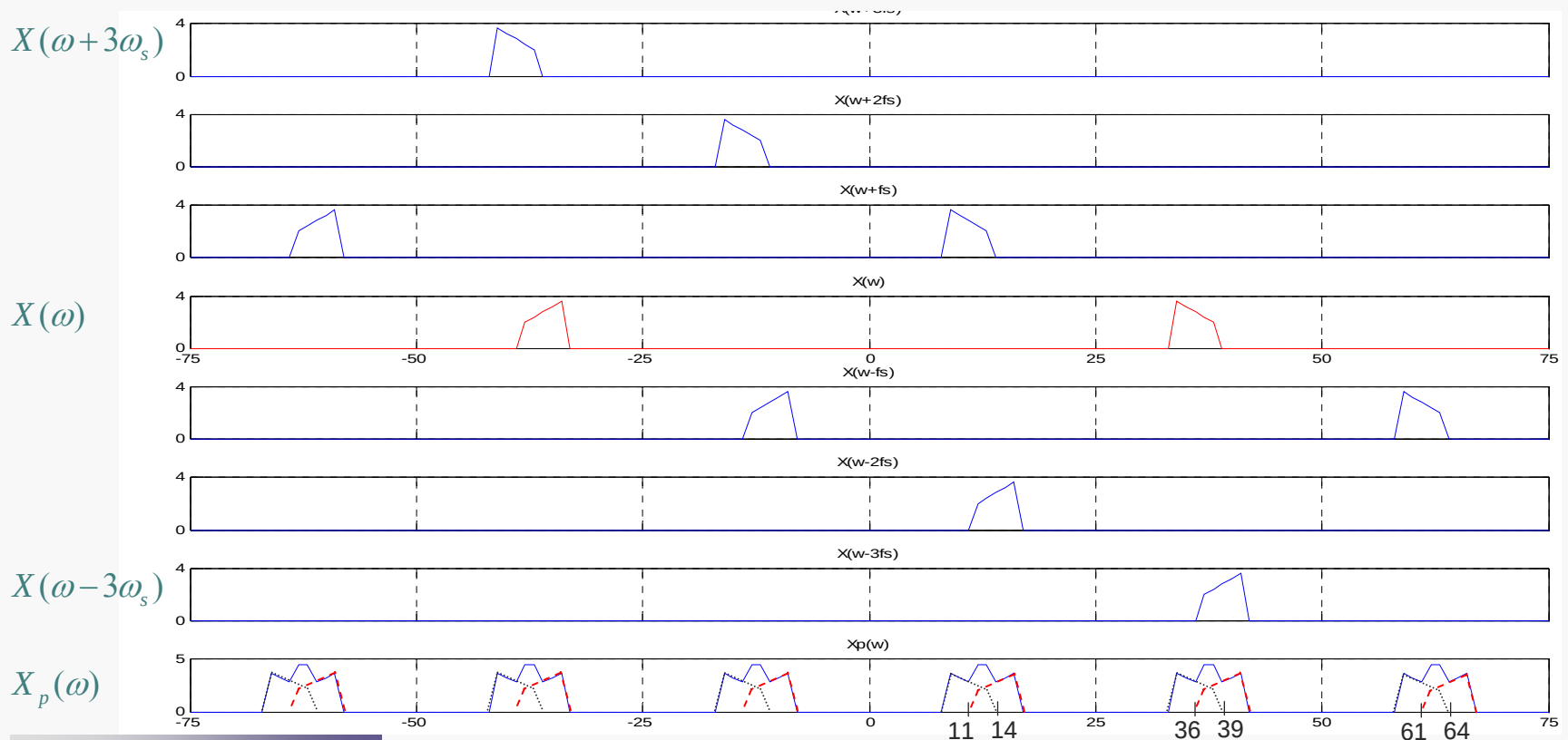
■ Sub sampling with 25 MHz $X_p(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s) \Big|_{\omega_s = 25 \text{ MHz}}$



Subsampling DTV signal (2)

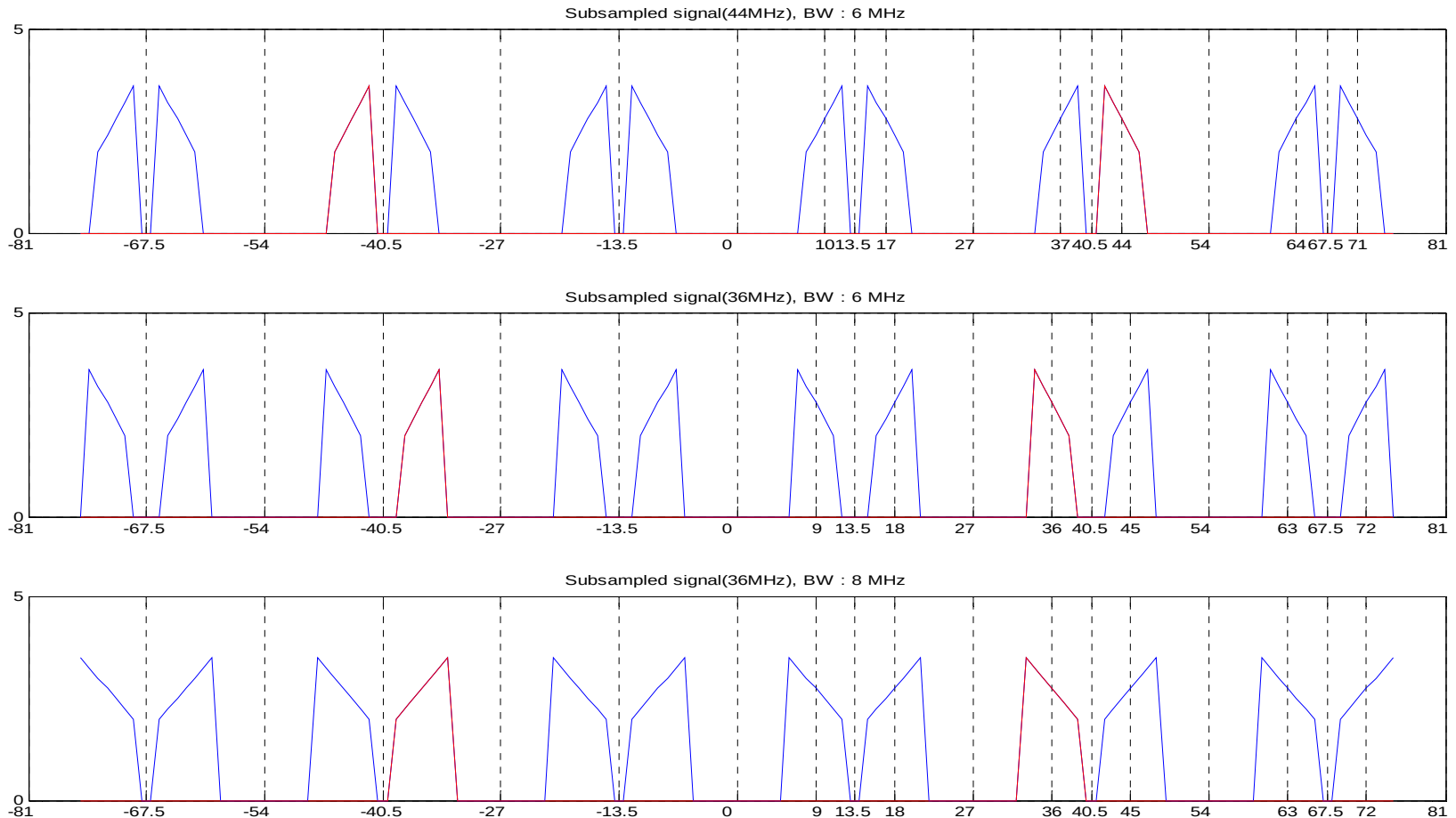
◇ Case 2 : OFDM (36MHz $\pm$ 3MHz)

■ Signal altered : subsampling freq. must be changed



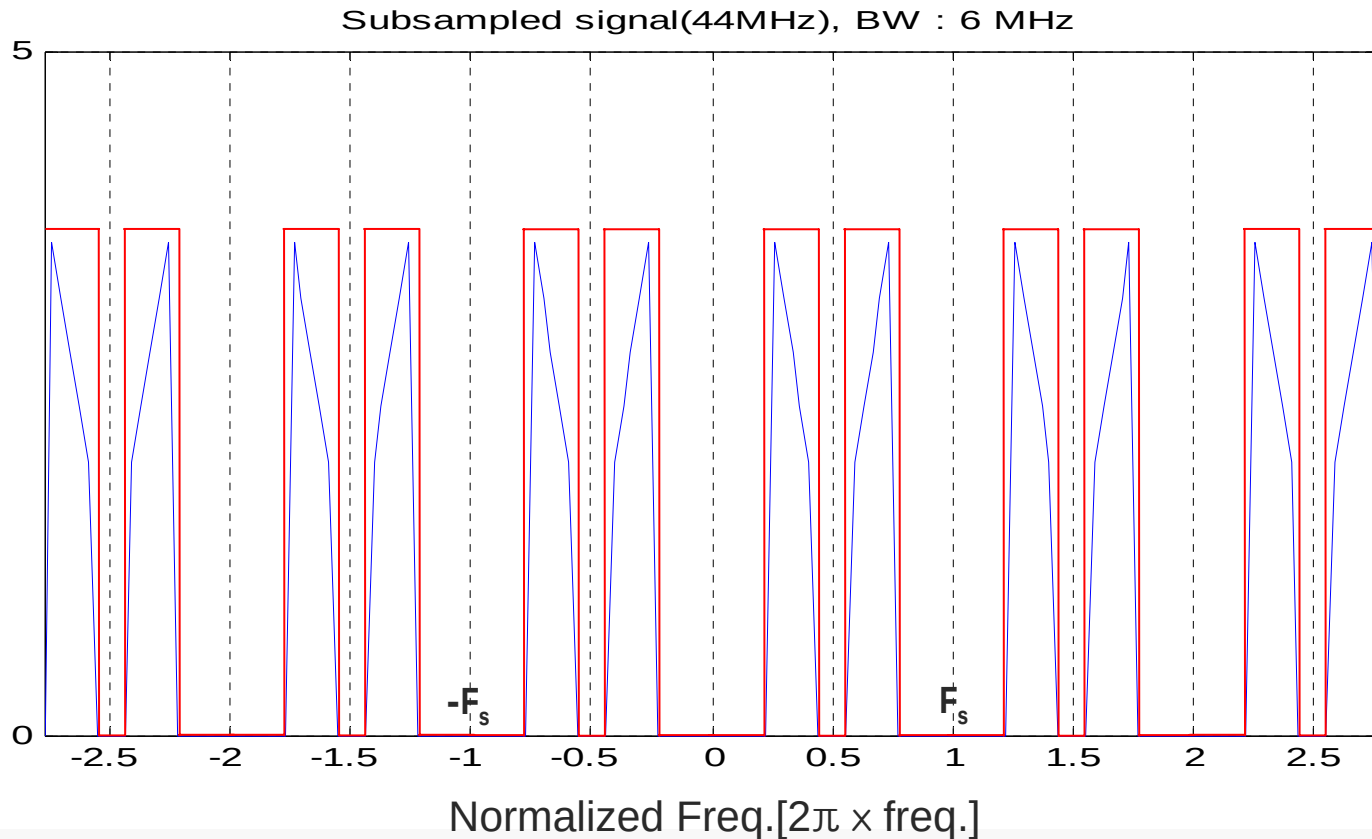
Suggesting Subsample Rate

◆ Subsampling with 27 MHz (0.5 MHz transition band)



Filter Design

◇ Graphical view of target bandpass filter



Filter Design : Case 1

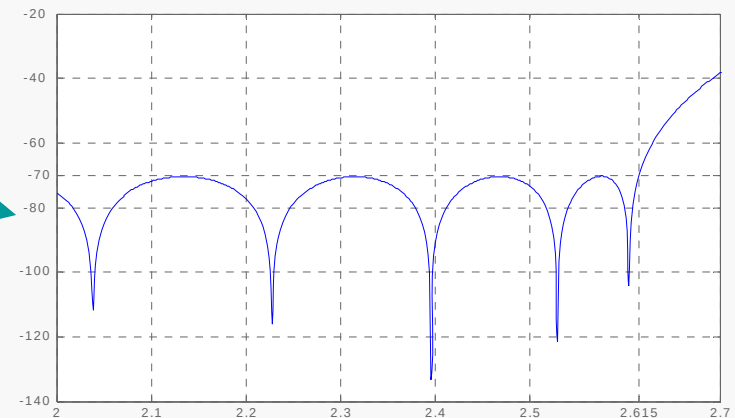
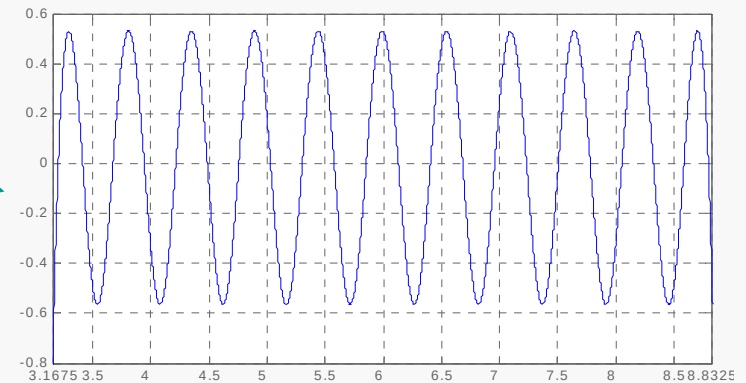
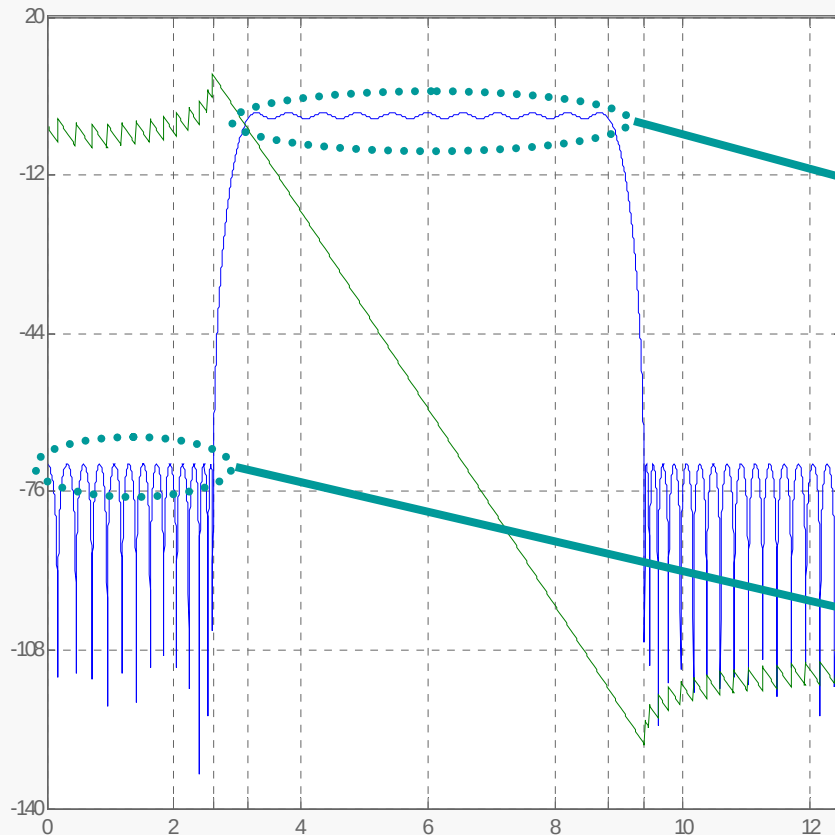
◇ Case 1 : Transition Band : 500 KHz ($\omega_s - \omega_c$)

■ VSB/QAM : 44 MHz $\pm$ 3MHz

■ Center Freq. : 6 MHz

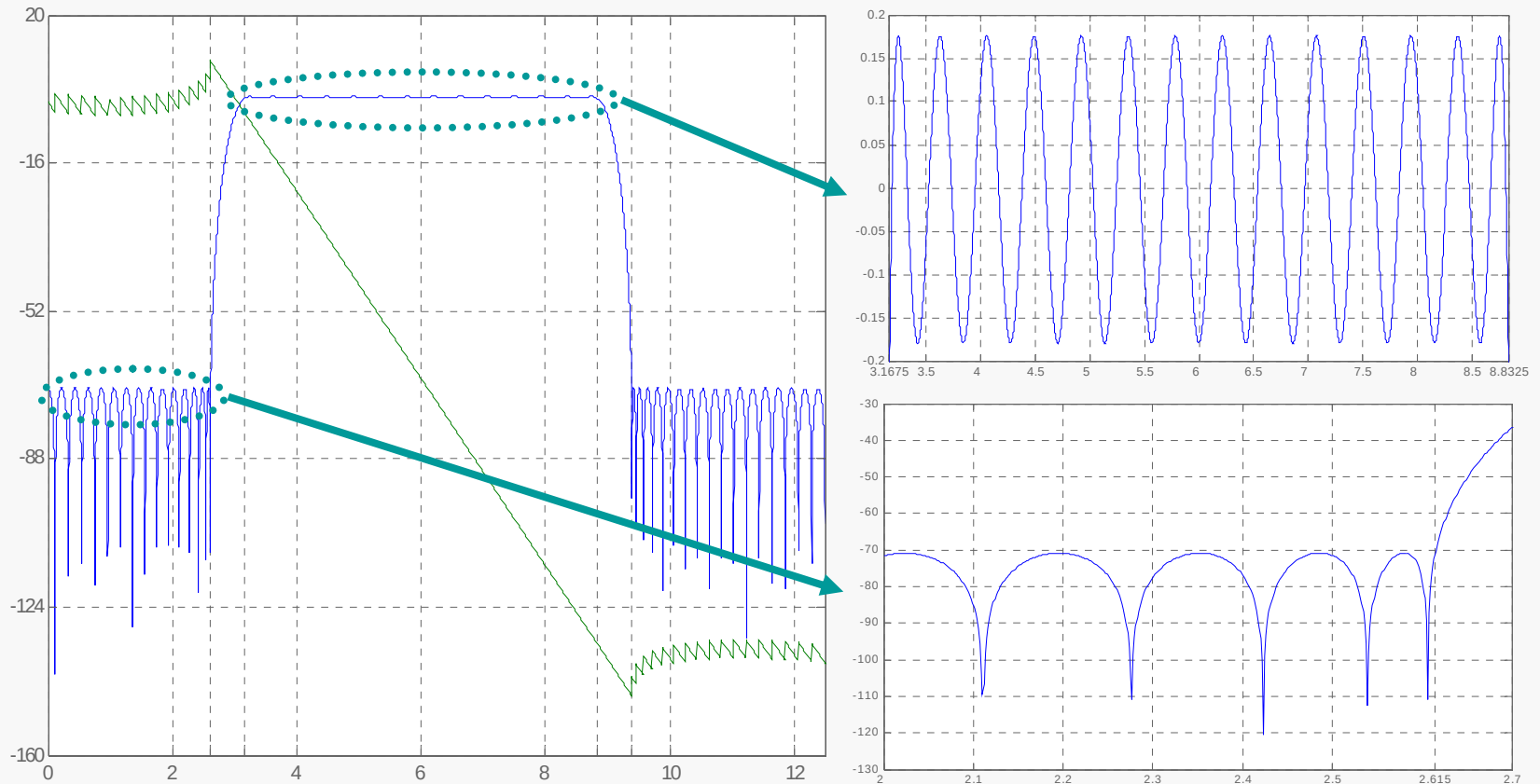
항목	목표	비고
Input Resolution	10 bit	
Clock frequency	25 MHz	
Passband	3.1675MHz ~ 8.8325 MHz	$\pm$ 167.5 KHz (0.1267, 0.3533)
Stopband	0~2.615 MHz 9.385~12.5 MHz	$\pm$ 385 KHz (0.1046, 0.3754)
Max pass-band ripple	$\pm$ 0.2 dB	
Min stop-band attenuation	70 dB	Output Resolution Required : 12 bit

FIR filter (105 tap)



◇ Passband ripple did not meet the spec.

FIR filter (124 tap)

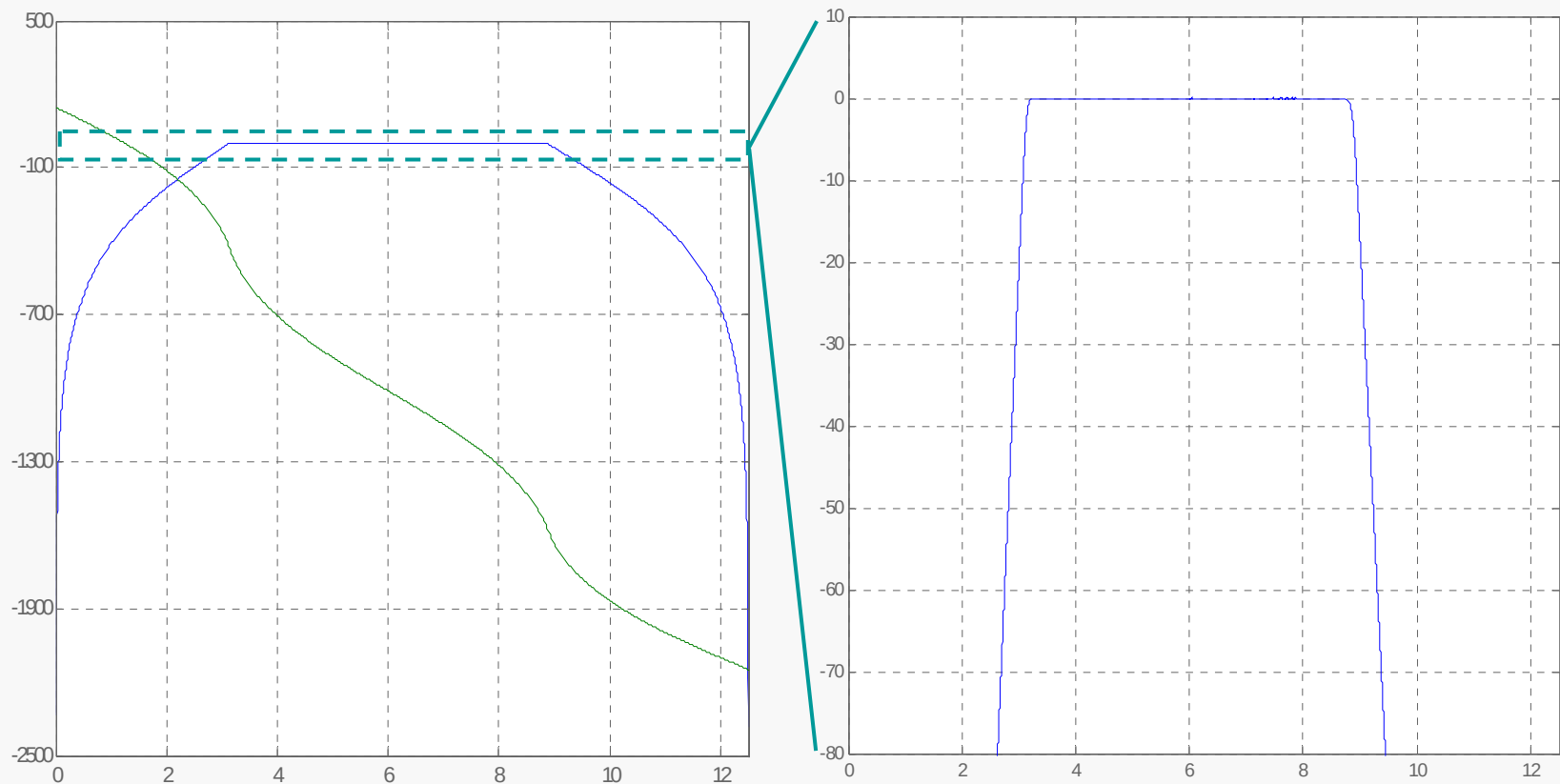


◆ Designed filter meets the spec.



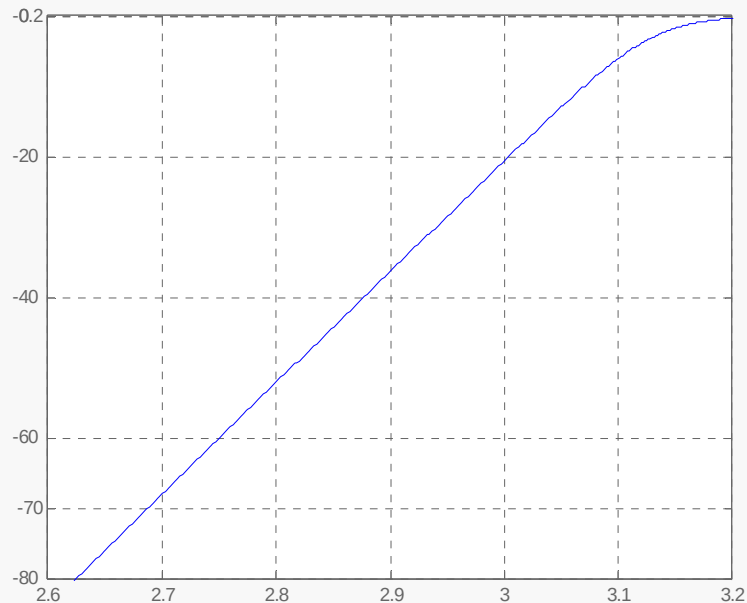
IIR : Butterworth (1)

◇ Magnitude response of filter (1)

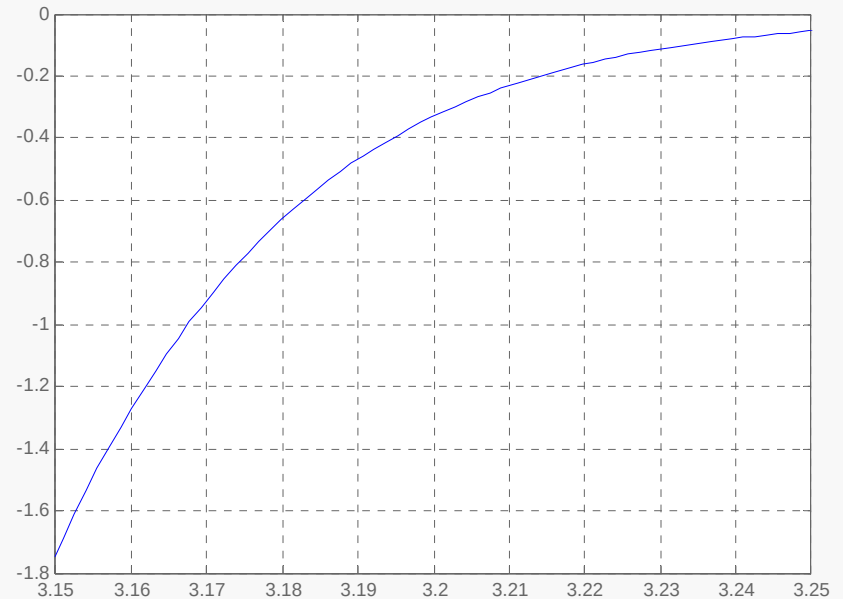


IIR : Butterworth (2)

◆ Magnitude response of filter (2)



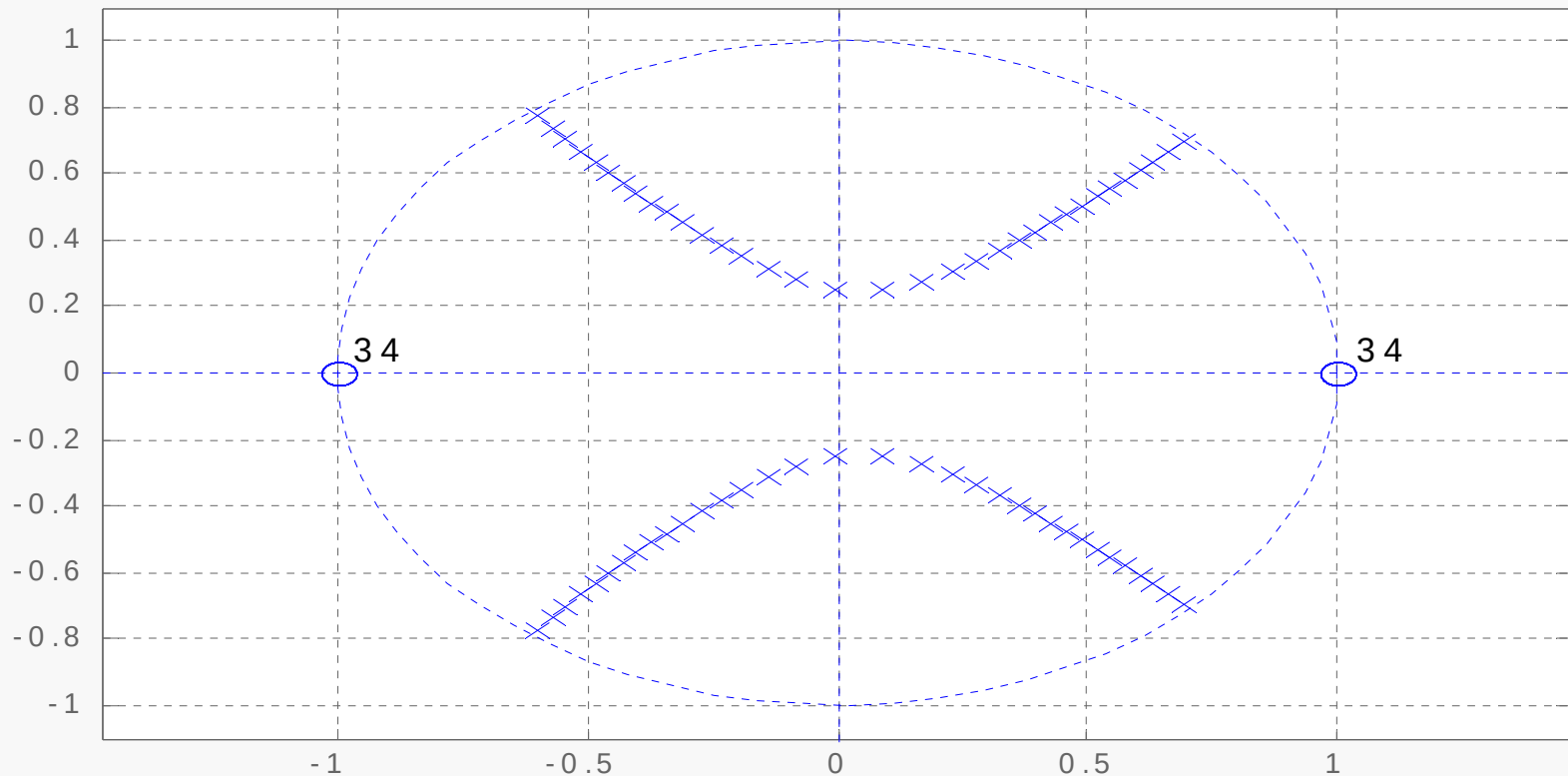
Stopband : -80 dB @2.615 MHz



Passband Edge

IIR : Butterworth (3)

◇ Pole/Zero Location

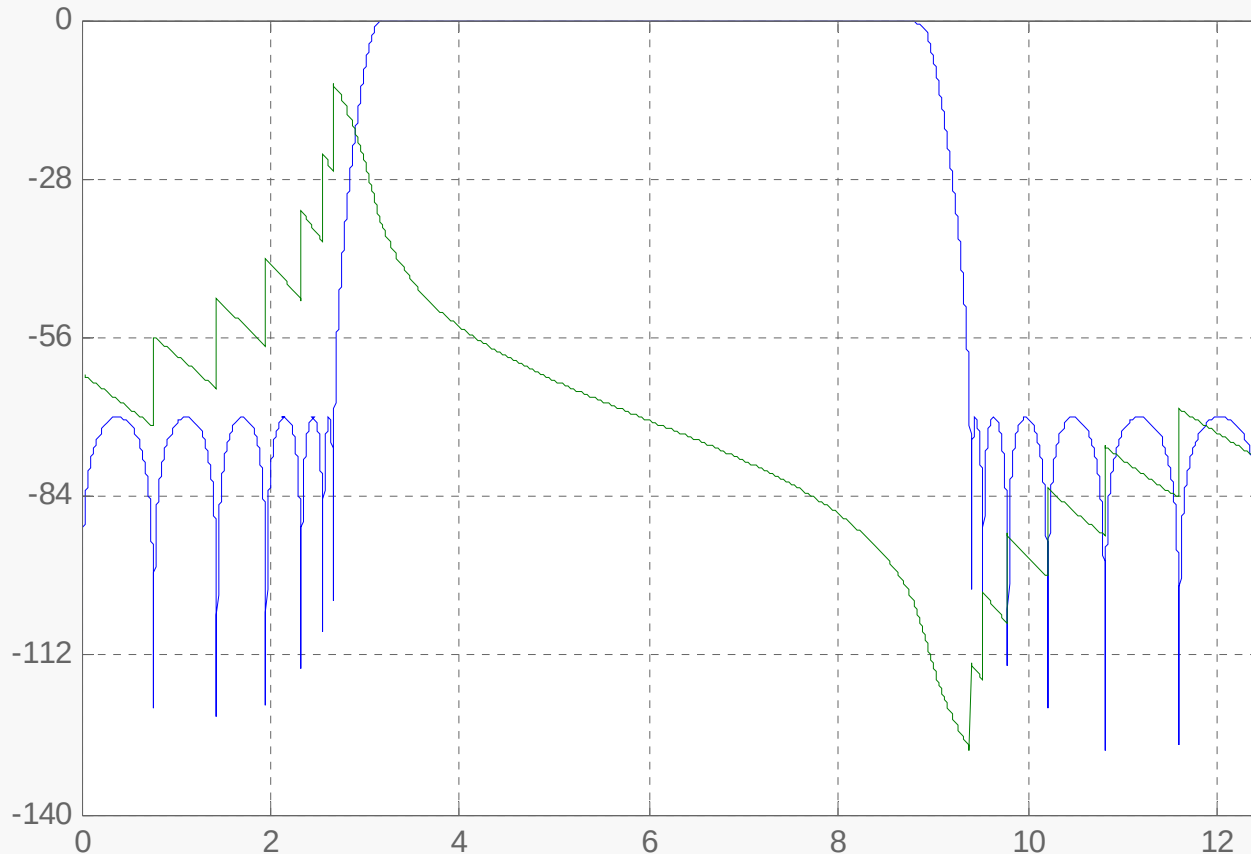


of poles : 68, # of zeros : 2



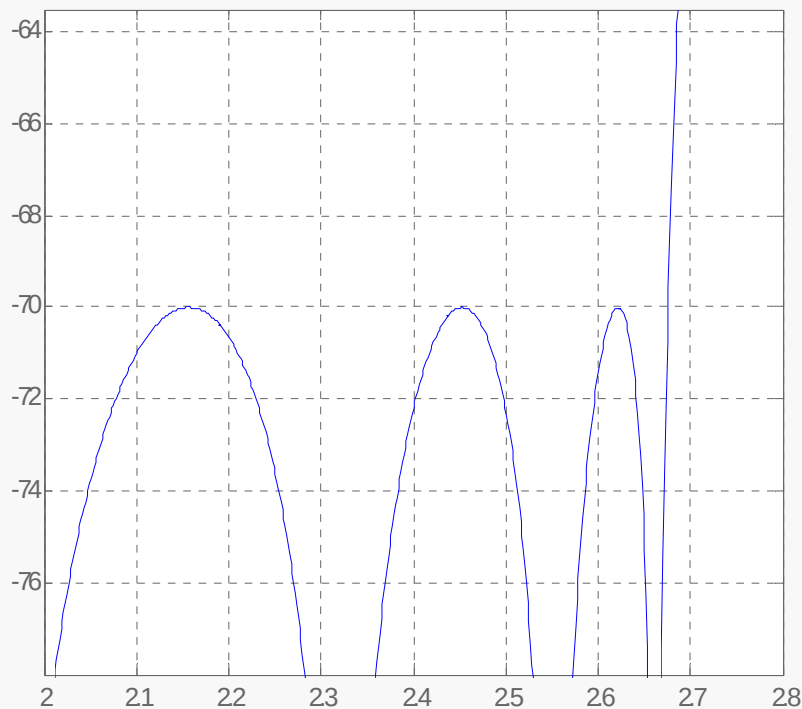
IIR : Chebyshev Type II (1)

◇ Magnitude response of filter (1)

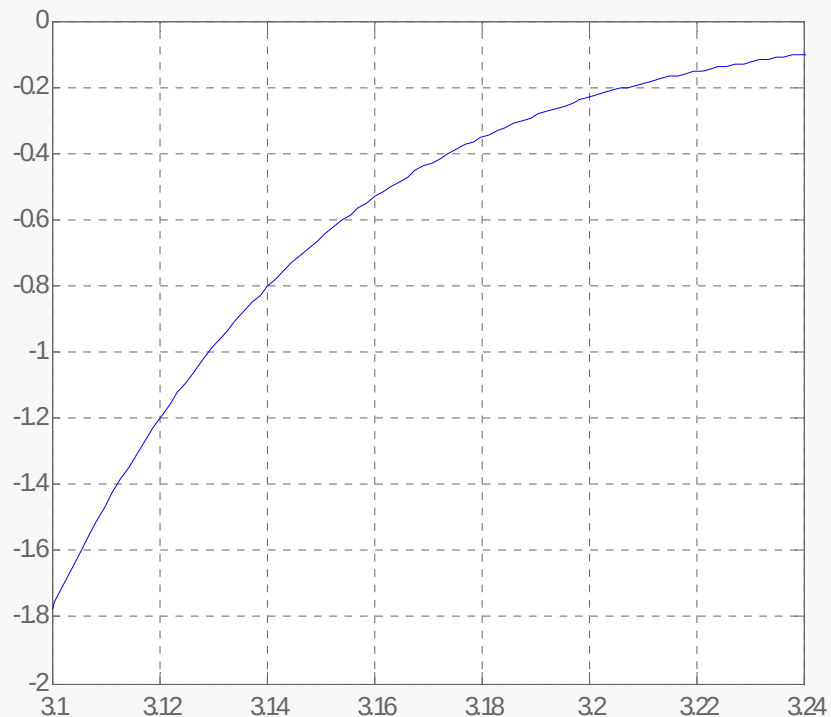


IIR : Chebyshev Type II (2)

◆ Magnitude response of filter (2)



Minimum Stopband Attenuation
: 70 dB

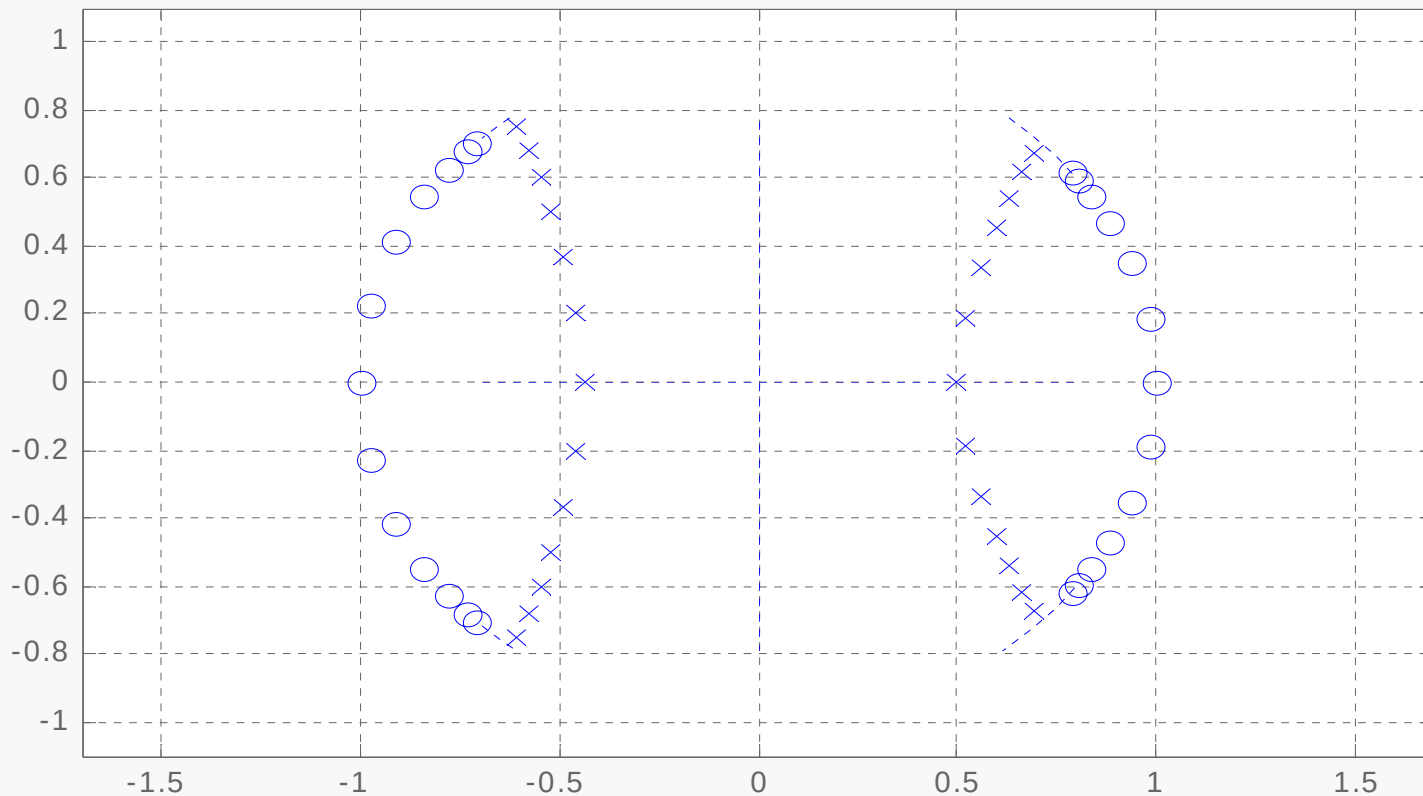


Passband Edge



IIR : Chebyshev Type II (3)

◇ Pole/Zero Location



of poles : 26, # of zeros : 26



Summary of Filter Design : Case 1

◇ Simulation Results of Case 1 (Transition Band : 500 KHz)

	Complexity	Passband Ripple (0.2 dB)	Stopband Attenuation (70 dB)
FIR	105 tap	0.6 dB	70 dB
FIR	124 tap	0.2 dB	70 dB
IIR (Butterworth)	Pole : 68, Zero : 2	1 dB	70 dB
IIR (Chebyshev Type II)	Pole : 26, Zero : 26	0.4 dB	70 dB



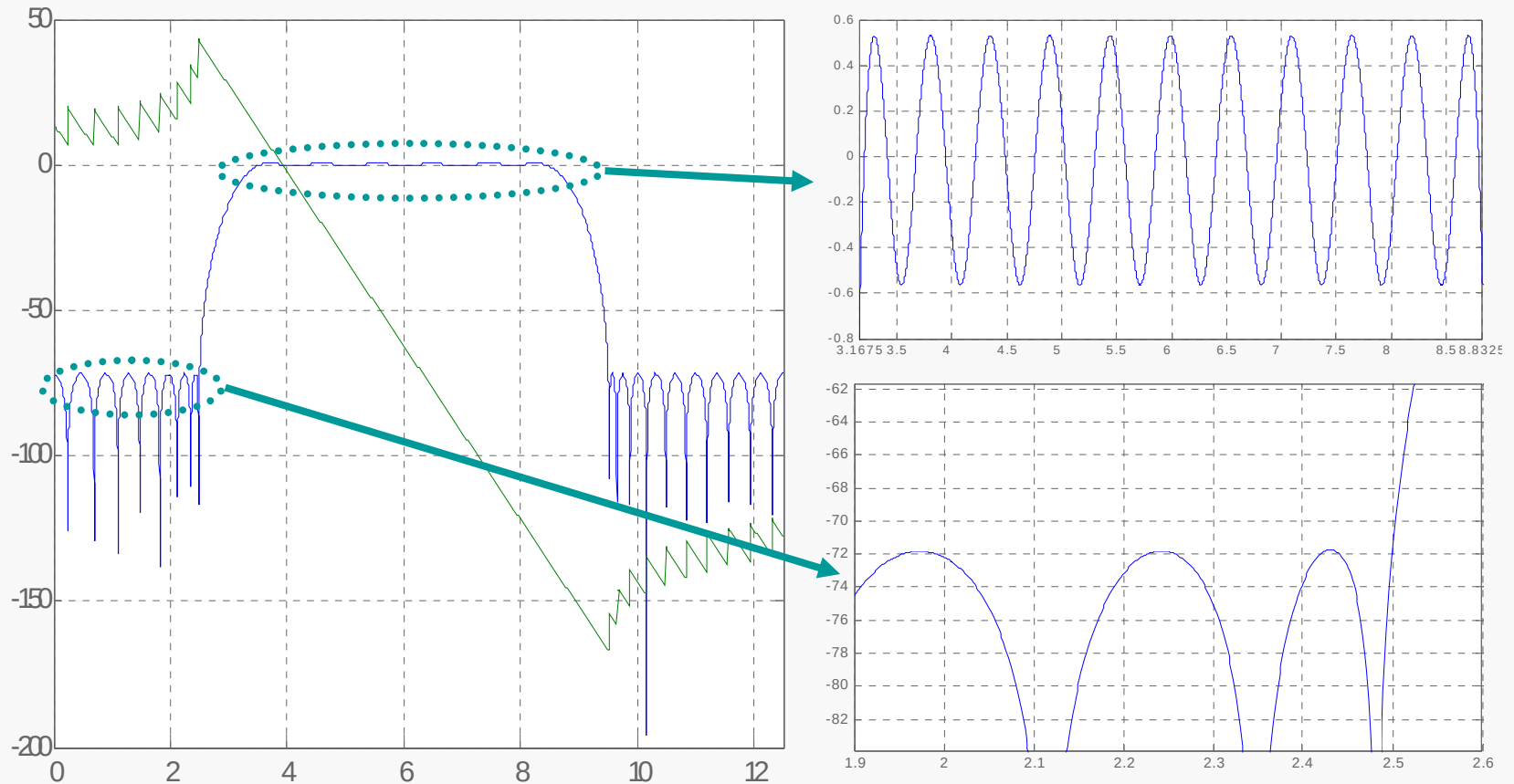
Filter Design : Case 2

◇ Case 2 Transition Band : 1 MHz ($\omega_s - \omega_c$)

항목	목표	비고
Input Resolution	10 bit	
Clock frequency	25 MHz	
Passband	3.5MHz ~ 8.5 MHz	± 500 KHz (0.14, 0.34)
Stopband	0~2.5 MHz 9.5~12.5 MHz	± 500 KHz (0.1, 0.38)
Max pass-band ripple	± 0.2 dB	
Min stop-band attenuation	70 dB	Output Resolution Required : 12 bit



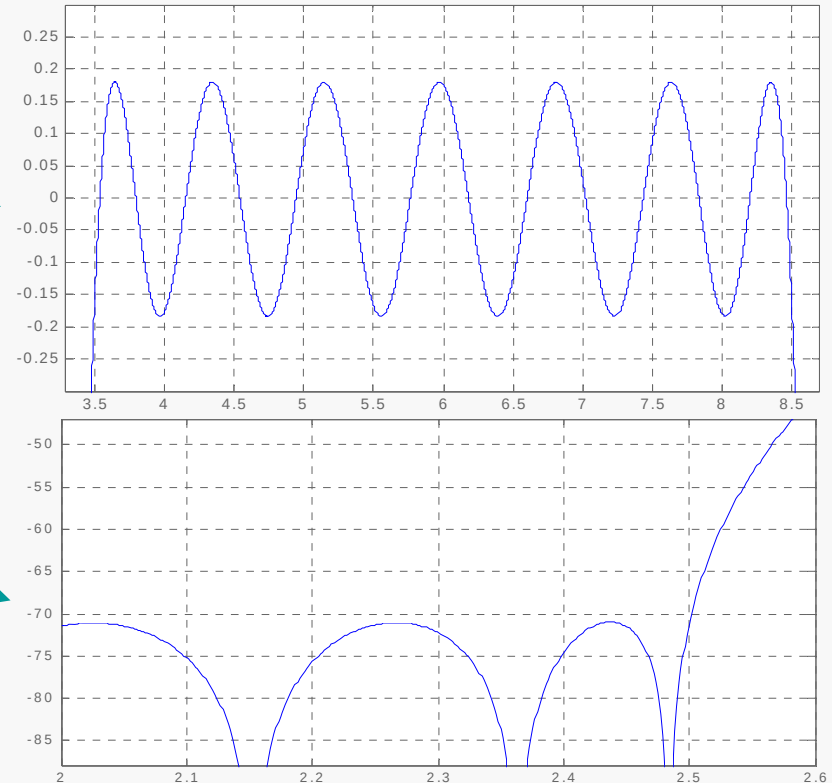
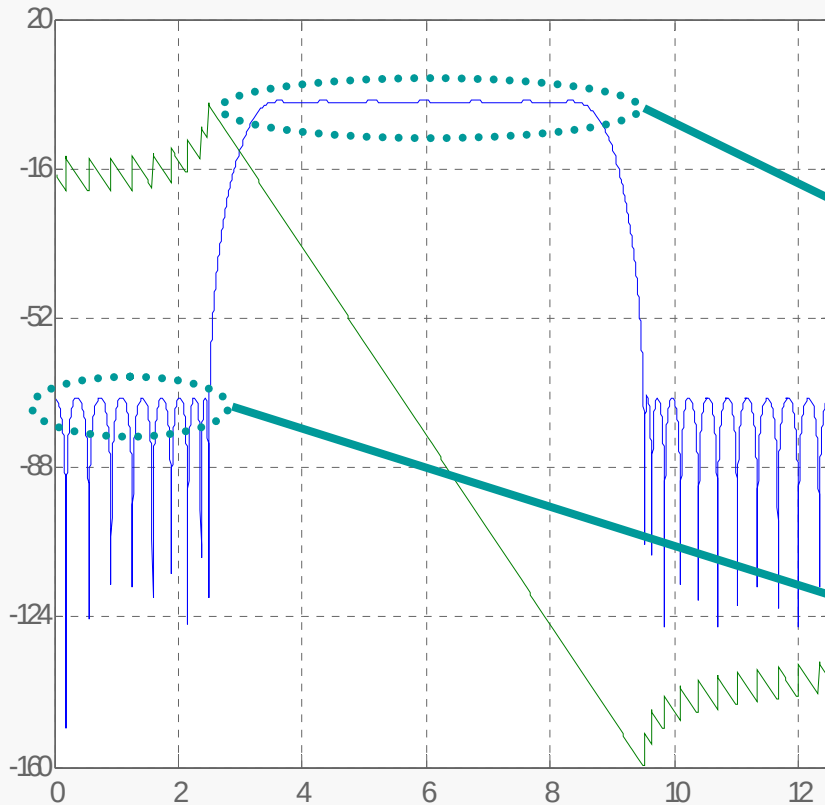
FIR filter (59 tap)



◇ Passband ripple did not meet the spec



FIR filter (71 tap)



◆ Designed filter meets the spec.



Summary of Filter Design: Case 2

◇ Simulation Result of Case 2 (Transition Band : 1 MHz)

	Complexity	Passband Ripple (0.2 dB)	Stopband Attenuation (70 dB)
FIR	59 tap	0.5 dB	70 dB
FIR	71 tap	0.2 dB	70 dB
IIR (Butterworth)	Pole : 68, Zero : 2	0.1 dB	70 dB
IIR (Chebyshev Type II)	Pole : 26, Zero : 26	0.1 dB	70 dB

